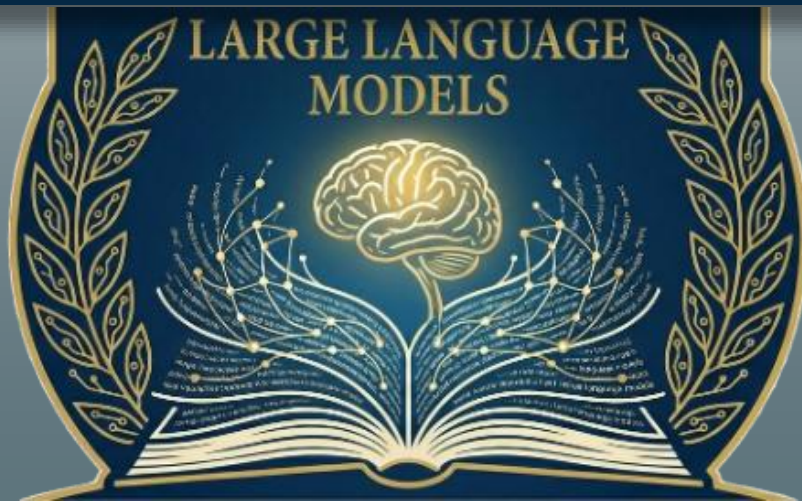


Data and Tokenization



Mohammad Javad Dousti

Sources:

- Princeton COS 597G course slides by A. N. Rubungo and T. Banerjee
- Mohit Iyyer's Advanced NLP's slides from UMass.

A famous tweet



Big Data Borat
@BigDataBorat



In Data Science, 80% of time spent prepare data, 20% of time spent complain about need for prepare data.

6:17 AM · Feb 27, 2013 · Twitter Web Client

544 Retweets **24** Quote Tweets **394** Likes

What are the goals of pre-training?

- ❑ The goals of pre-training are to get the language model to:
 - learn the structure of natural language
 - learn humans' understanding of the world (as encoded in the training data).
- ❑ Ideal data for pre-training:
 - Data that is high-quality, clean, and diverse.
 - Books, Wikipedia, news, scientific papers, etc.

Pre-training data reality

- More data is almost always better.

RoBERTa_{Large} Pretrained with Different Data Setups[10].

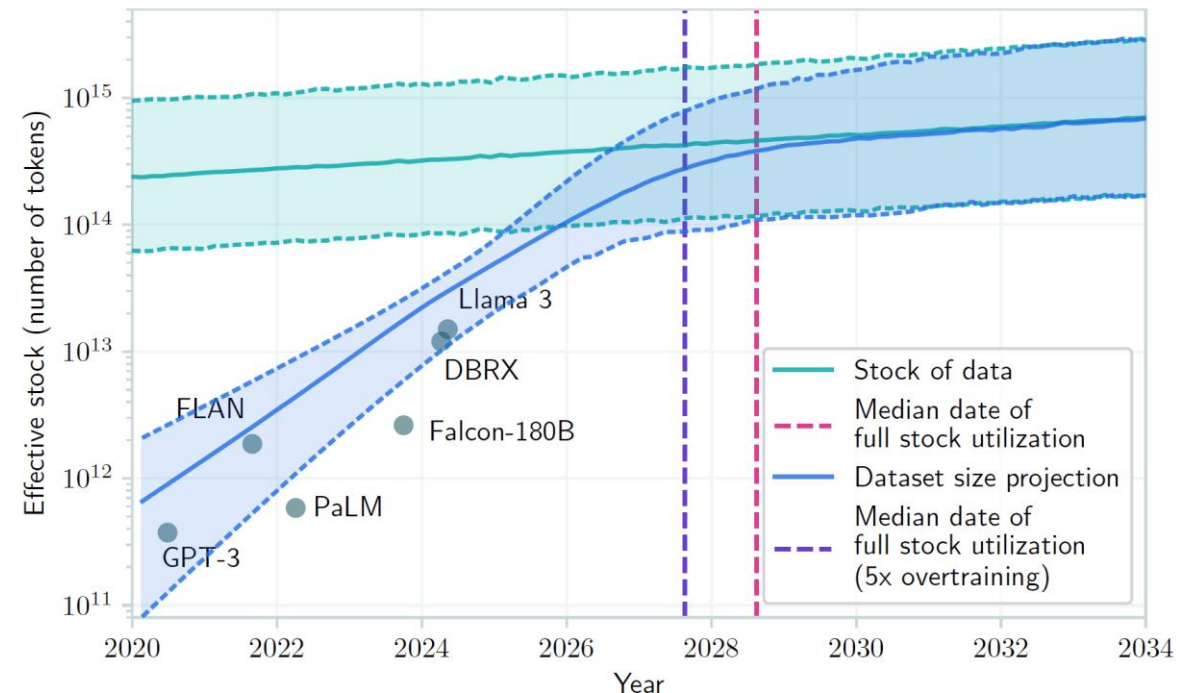
Pretrain Setups	Data Size	Batch Size	Steps	MNLI-m (ACC)	SQuAD 1.1 (F1)
Wiki + Books	16GB	8K	100K	89.0	93.6
+Additional Data	160GB	8K	100K	89.3	94.0
+Pretrain More Steps	160GB	8K	300K	90.0	94.4
+Even More Steps	160GB	8K	500K	90.2	94.6

- Same model, same pretraining steps, more data helps
- With more data more pretraining steps also help
- Empirical gains more than many “fancier” improvements

Source: Li, et al. "RoBERTa: A robustly optimized BeRT pretraining approach." arXiv preprint arXiv:1907.11692. 2019.

Pre-training Data Reality

- ❑ High quality data eventually runs out.
- ❑ In practice, the web is the most viable option for data collection.
 - In the digital era, this is the go-to place for general domain human knowledge.
 - It is massive and unlikely to grow slower than computing resources.
 - Publicly available.



V. Pablo et al. "Will we run out of data? Limits of LLM scaling based on human-generated data", arXiv 2024.

Pre-training Data Reality

- ❑ Web data is plentiful, but can be challenging to work with.
 - Copyright and usage constraints can get extremely complicated
 - Data is noisy, dirty, and biased
 - Data is contaminated with auto-generated text
 - Not just from LLM usage, but also tons of templated text.

The Web Data Pipeline

1. Content is posted to the web.
2. Webcrawlers identify and download a portion of this content.
3. The data is filtered and cleaned.

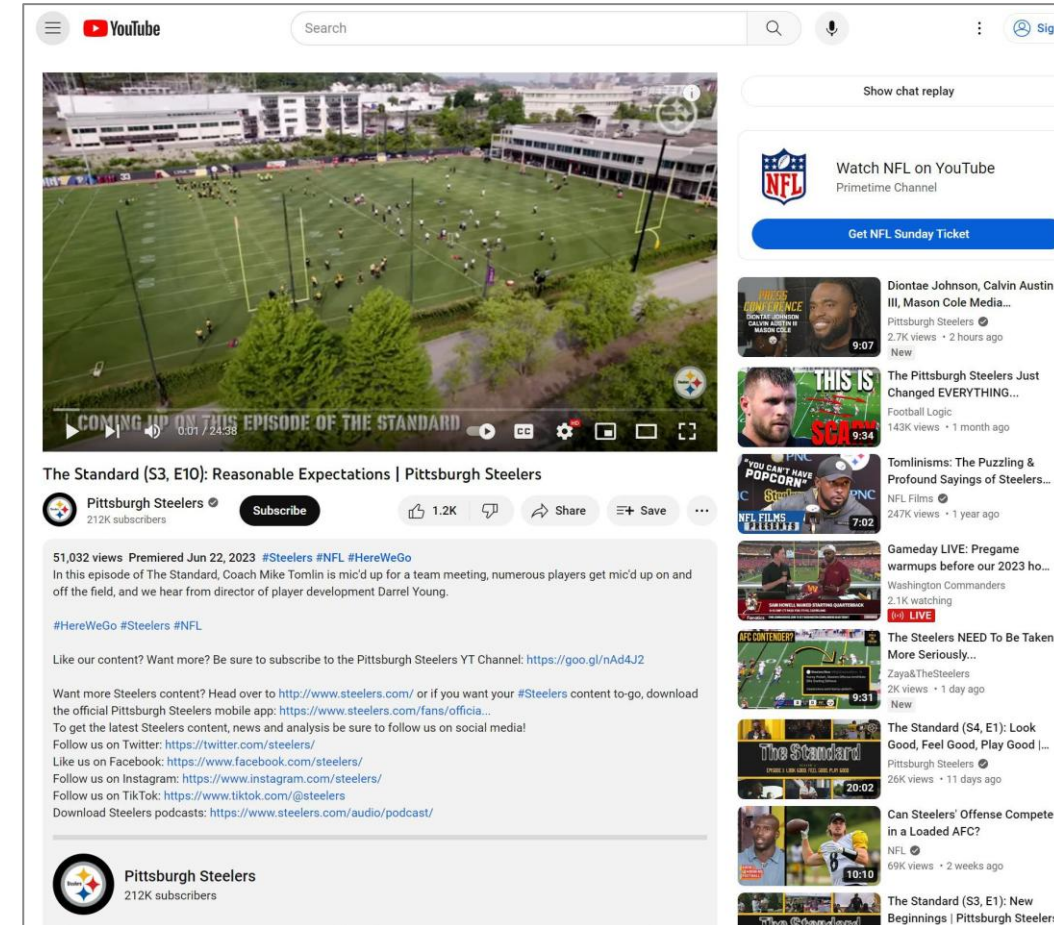
The Web Data Pipeline

- 1. Content is posted to the web.**
2. Webcrawlers identify and download a portion of this content.
3. The data is filtered and cleaned.

1. Content is posted to the internet.

❑ Challenges

- Biases in what's available:
 - Recency bias
 - Demographic biases
 - Language biases
- Web is much more dynamic than static HTML pages
 - CSS, JavaScript, interactivity, etc.
 - Responsive design
 - Many HTML pages involves 20+ secondary URLs, iframes, etc.
- What counts as content?
 - Ads, recommendation, navigation, etc.
 - Multimedia: images, videos, tables, etc.
 - Spam



1. Content is posted to the internet.

- Content extraction from webpages is a well-studied problem in industry.
 - Can be very engineering and resource heavy to do well.
 - Existing toolkits is a strategic advantage of some proprietary LLMs



A Content Extraction Pipeline from Bing Used for ClueWeb22

The Web Data Pipeline

1. Content is posted to the web.
- 2. Webcrawlers identify and download a portion of this content.**
3. The data is filtered and cleaned.

2. Webcrawlers identify and download a portion of this content.

□ General Idea

- Start with a set of seed websites
- Explore outward by following all hyperlinks on the webpage.
- Systematically download each webpage and extract the raw text.



□ Challenges

- How to harvest a large number of seed URLs efficiently
- How to select “high quality” URLs and skip over “bad” URLs
 - Some cases are clear cut: spammy, unsafe, etc.
 - Some are hard to detect or up to debate: certain biases.
- How to keep the crawl up-to-date
- Given a fixed compute budget each month, is it better to crawl new webpages, or recrawl old ones that might’ve changed?

The Web Data Pipeline

1. Content is posted to the web.
2. Webcrawlers identify and download a portion of this content.
- 3. The data is filtered and cleaned.**

3. The data is filtered and cleaned.

- ❑ Remove noisy, spammy, templated, and fragmented texts
 - These portions lack the content needed to meet pretraining goals.
- ❑ Select higher quality texts from a massive candidate pool
 - Given a limited pretraining compute budget, we'd like pretrain on better texts
- ❑ Avoid toxic and biased content
 - NSFW content
 - Texts with strong biases

3. The data is filtered and cleaned.

- ❑ Methods to identify high-quality content:
 - Rule-based heuristics for quality
 - Proximity to known high-quality indicators
 - E.g. a website that was highly upvoted on Reddit, or a website that was included as a reference on Wikipedia
 - Classifiers
 - Trained quality and toxicity classifiers are common.
 - E.g. train a classifier with Wikipedia as the positive examples and random web pages as the negative examples

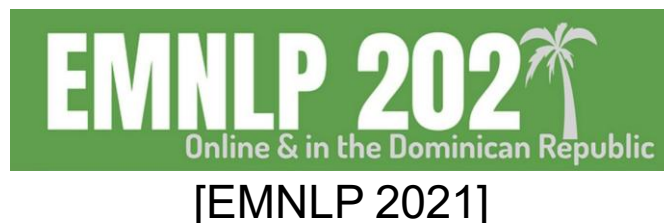
3. The data is filtered and cleaned.

❑ Challenges

- At what granularity should filtering be performed?
 - Word-level, sentence-level, paragraph-level, document level?
- What constitutes “high quality” or “non-toxic”?
- Are our filters/classifiers multilingual? Even within English, do they treat all groups equally?
- It is very expensive to ablate pre-training dataset decisions.

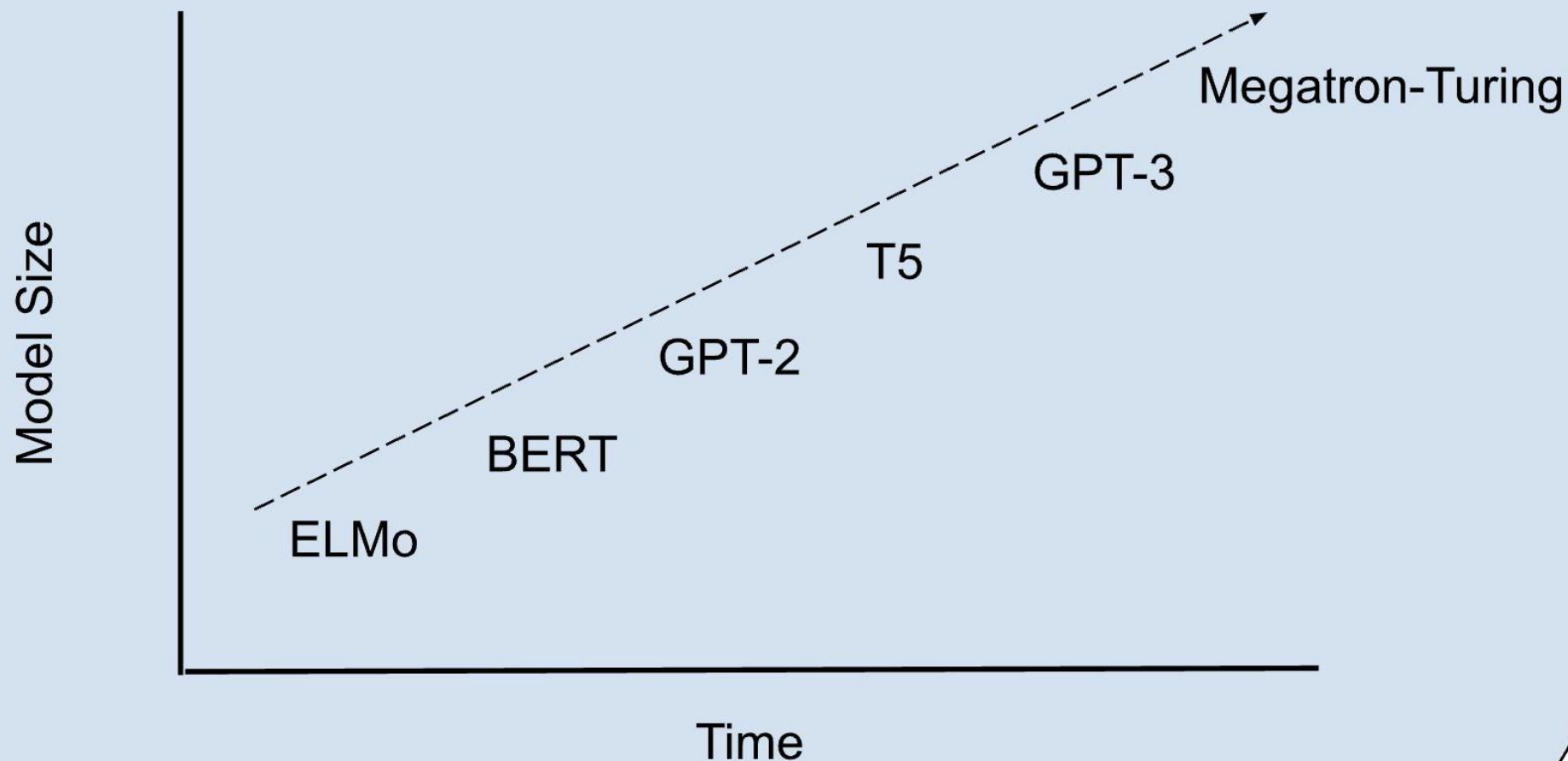
Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus

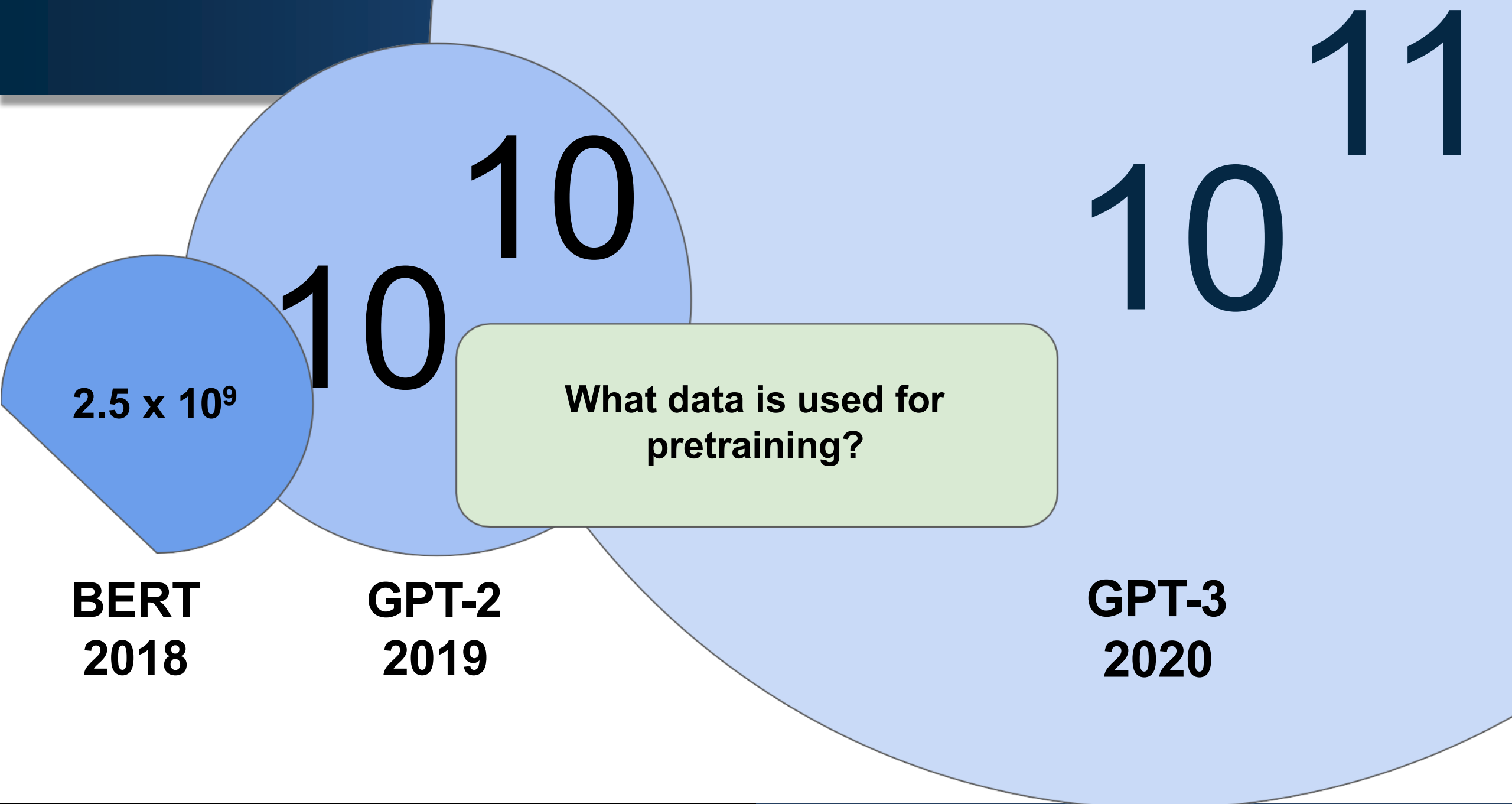
Jesse Dodge♣ Maarten Sap♣♡ Ana Marasović♣♡ William Agnew♦♡
Gabriel Ilharco♡ Dirk Groeneveld♣ Margaret Mitchell♠ Matt Gardner♣
♡Paul G. Allen School of Computer Science & Engineering, University of Washington
♠Hugging Face
♣Allen Institute for Artificial Intelligence



Some slides adapted from [Jesse Dodge's talk](#) @ EMNLP 2021

Pretrained language models





Web-scale text



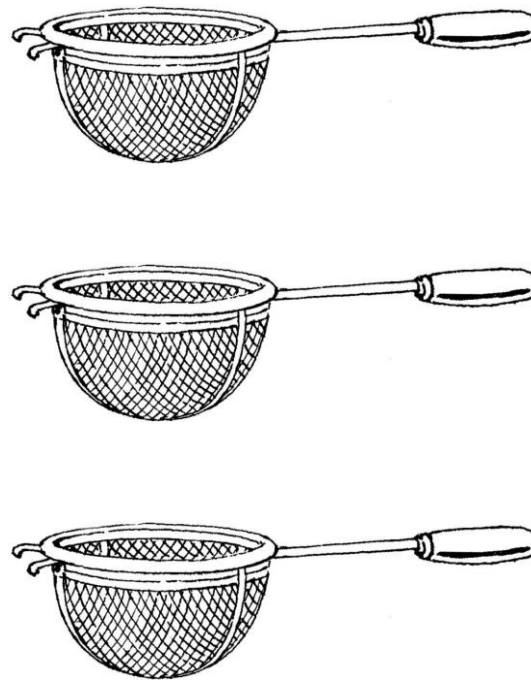
100 petabytes

Google search index

2.5 petabytes an hour

Data generated by Walmart

Web-scale text



Colossal Clean Crawled Corpus (C4)

Created to train T5 (Raffel et al. 2019)

April 2019 snapshot of **Common Crawl** (1.4 trillion tokens)

Remove lines without terminal punctuation mark, code (“{“), < 3 words

Remove docs with “lorem ipsum”, < 5 sentences, “bad words”

Used **langdetect** to filter out non-English text

Result: 305 GB of text (156 billion tokens)

- ❑ A nonprofit organization that crawls the web and freely provides its archives and datasets to the public.
- ❑ Common Crawl's web archive consists of *petabytes* of data collected since 2008.
- ❑ It completes crawls generally *every month*.
- ❑ CommonCrawl News (CC-News) was also a dataset containing news articles from news sites all over the world.

Crawl date	Size in TiB	Billions of pages	Comments
December 2023	454	3.35	Crawl conducted from November 28 to December 12, 2023
June 2023	390	3.1	Crawl conducted from May 27 to June 11, 2023
April 2023	400	3.1	Crawl conducted from March 20 to April 2, 2023
February 2023	400	3.15	Crawl conducted from January 26 to February 9, 2023
December 2022	420	3.35	Crawl conducted from November 26 to December 10, 2022
October 2022	380	3.15	Crawl conducted in September and October 2022

A few examples

Dataset	# documents	# tokens	size
C4.EN.NO CLEAN	1.1 billion	1.4 trillion	2.3 TB
C4.EN.NO BLOCKLIST	395 million	198 billion	380 GB
C4.EN	365 million	156 billion	305 GB

Introduced by Raffel et. al., 2020

Need for documentation

Task specific NLP datasets have best practices around documentation

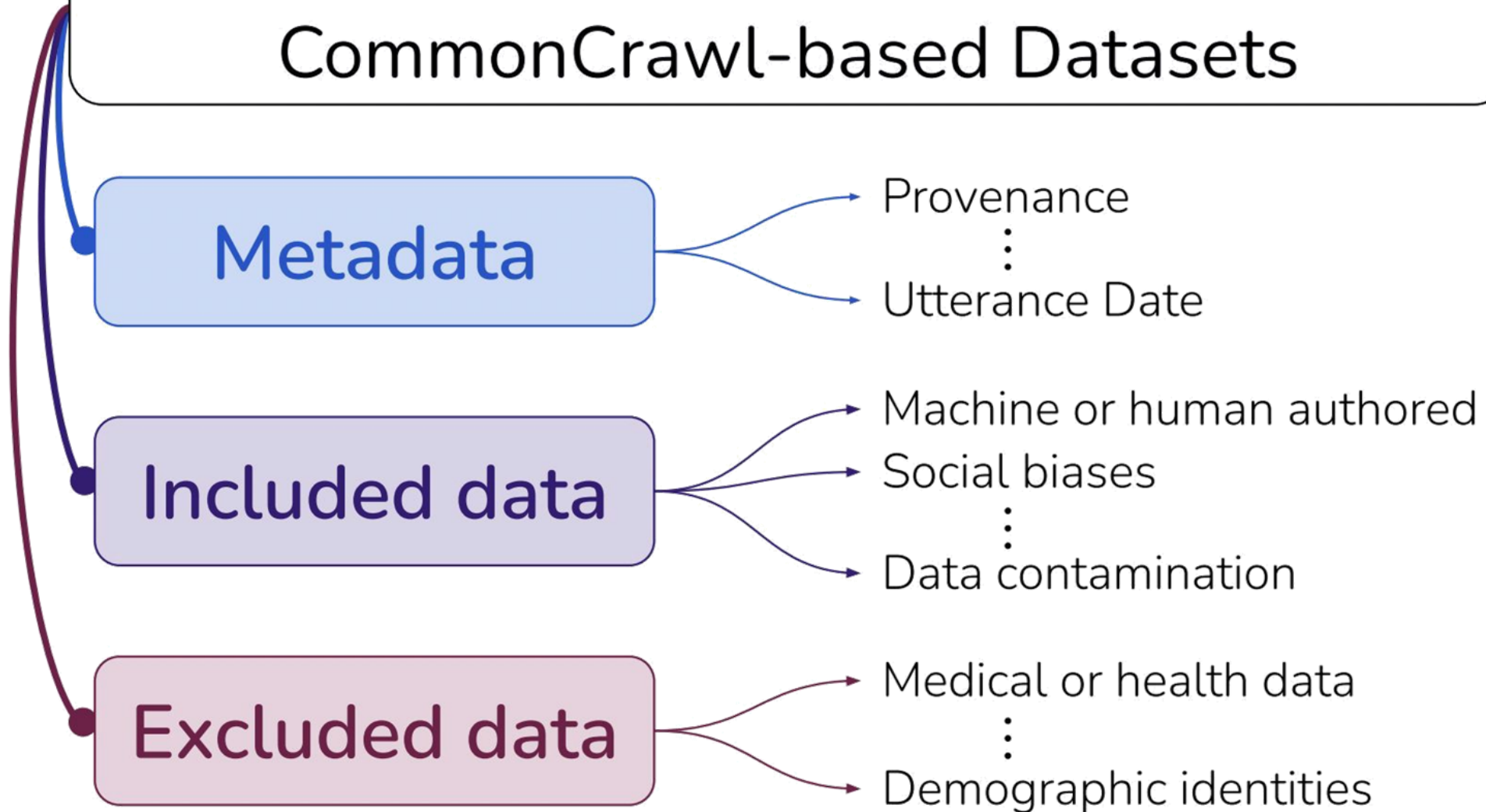
Applying these practices to massive datasets of unlabelled text is a challenge

Required information not available in web-crawled text

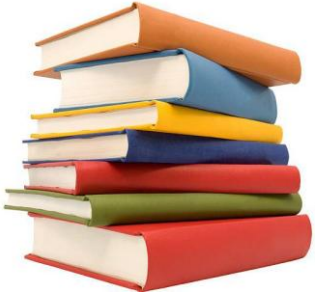
Thorough documentation typically not done

Leaves consumers of pretrained LMs in the dark about the influences of pretraining data on their systems

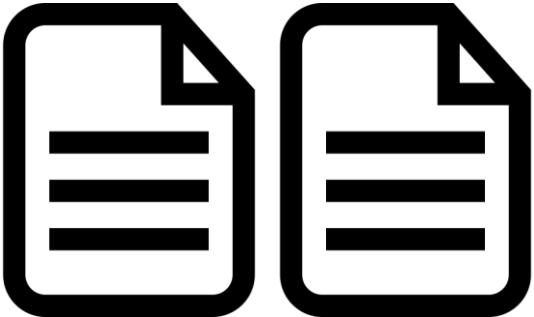
Documentation Levels for CommonCrawl-based Datasets



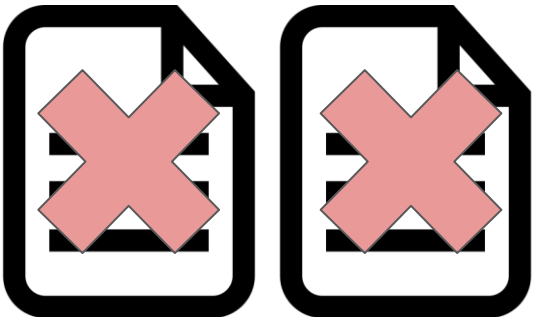
Documentation levels



Metadata



Included data



Excluded data

Corpus level statistics: Metadata

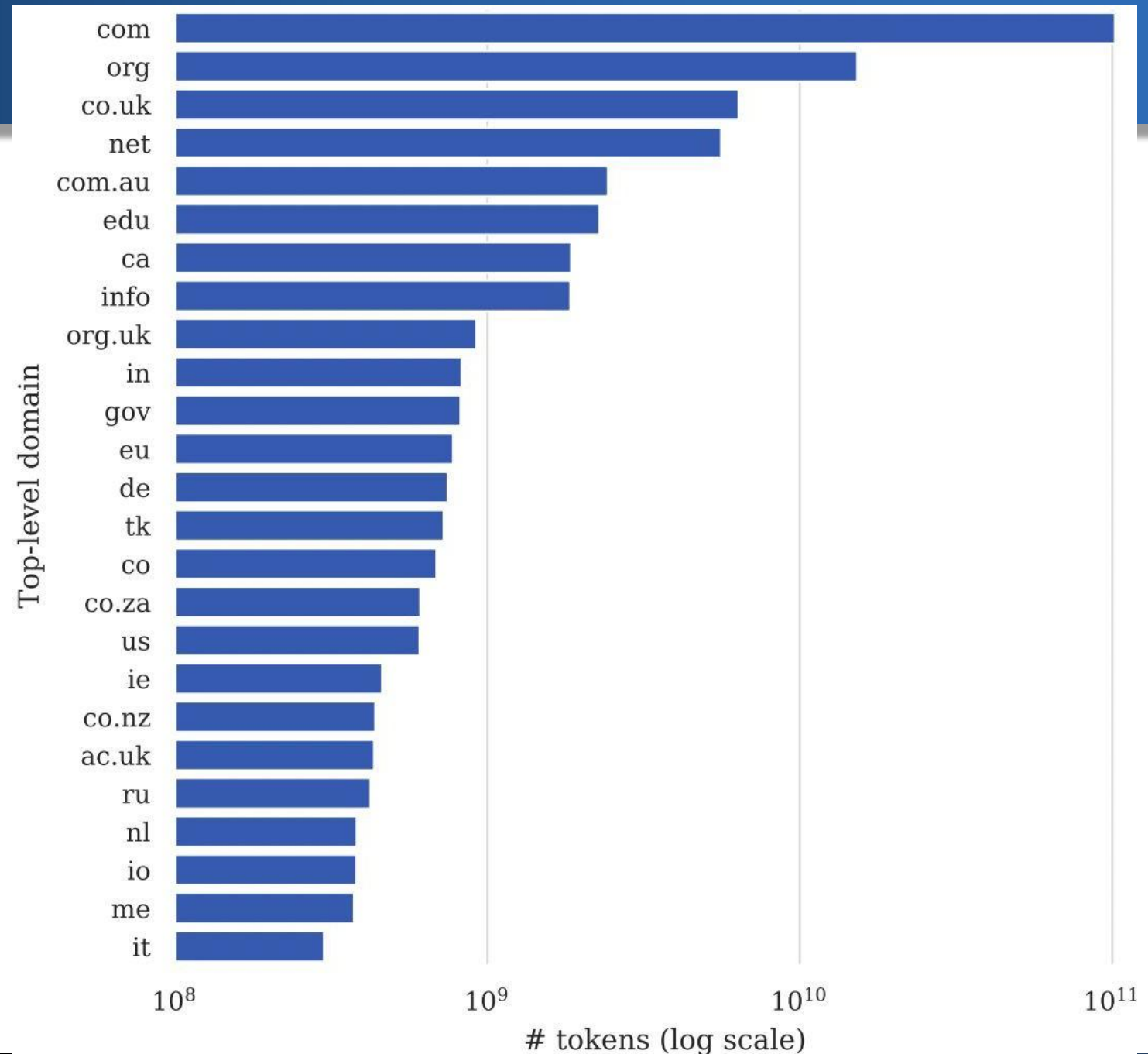
Internet domains and websites

Utterance data

Geolocation

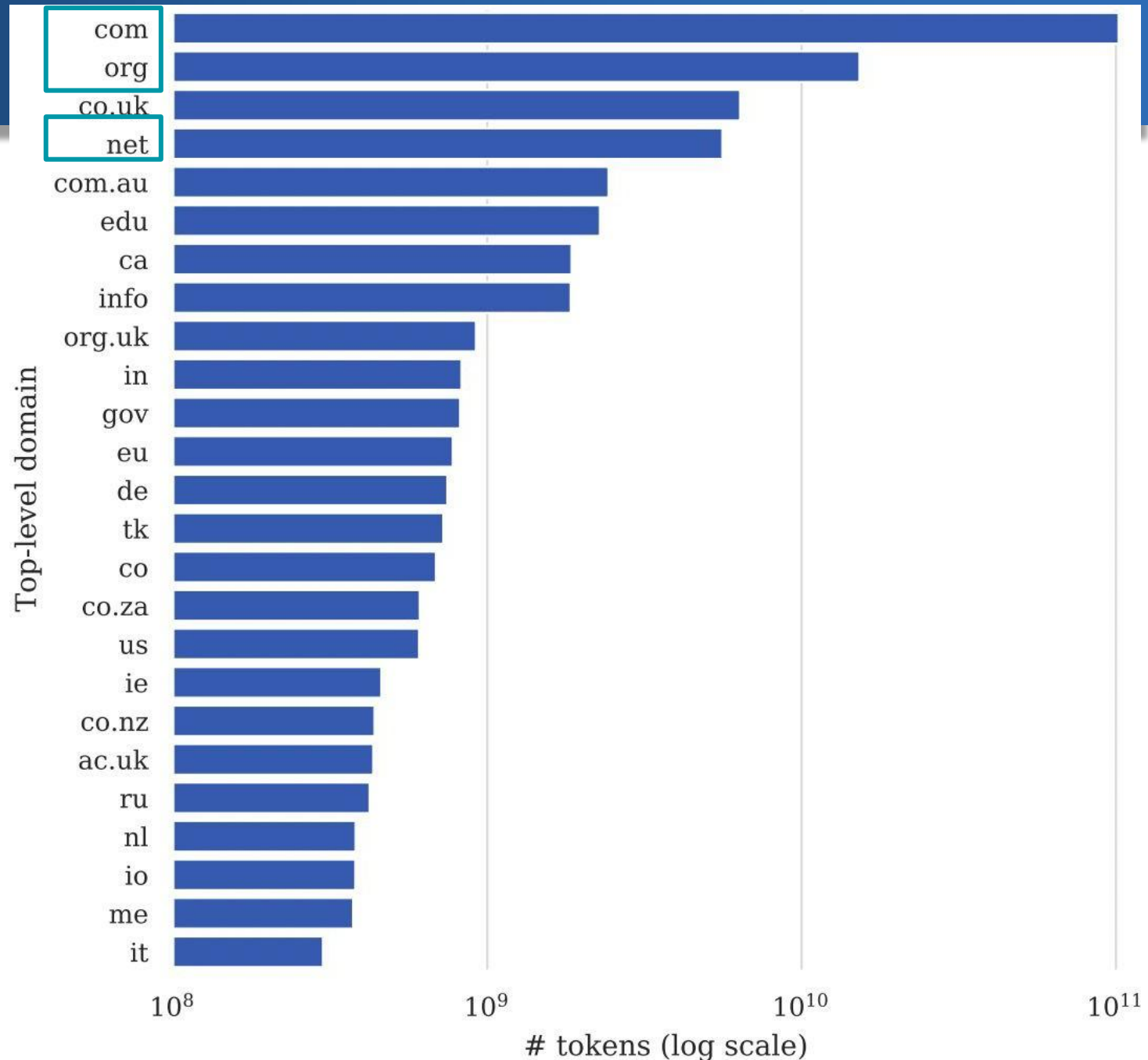
Metadata: Internet domains

- ❑ Top-level domains by number of tokens
- ❑ Tokens measured based on SpaCy tokenizer



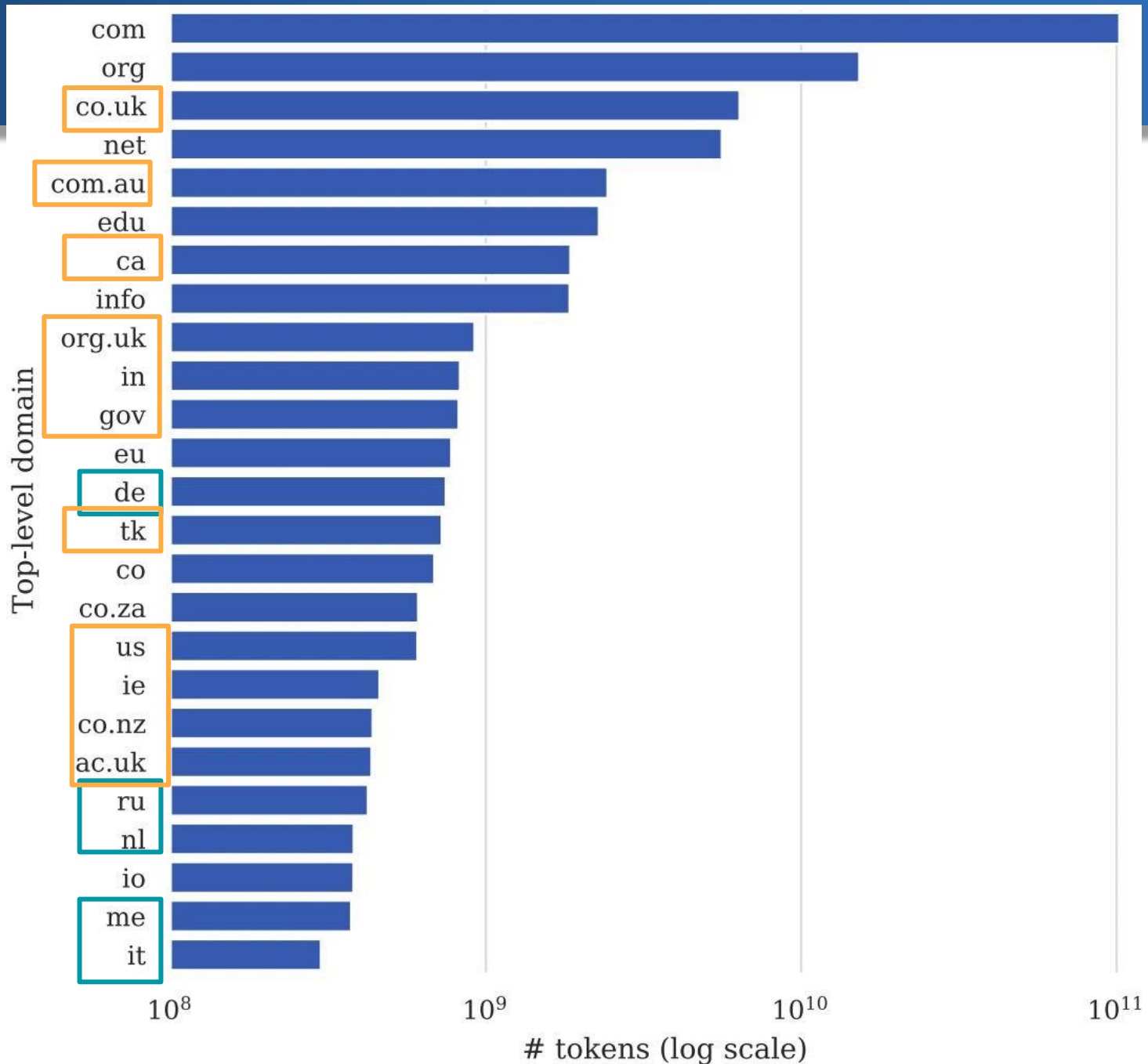
Metadata: Internet domains

- Top-level domains by number of tokens



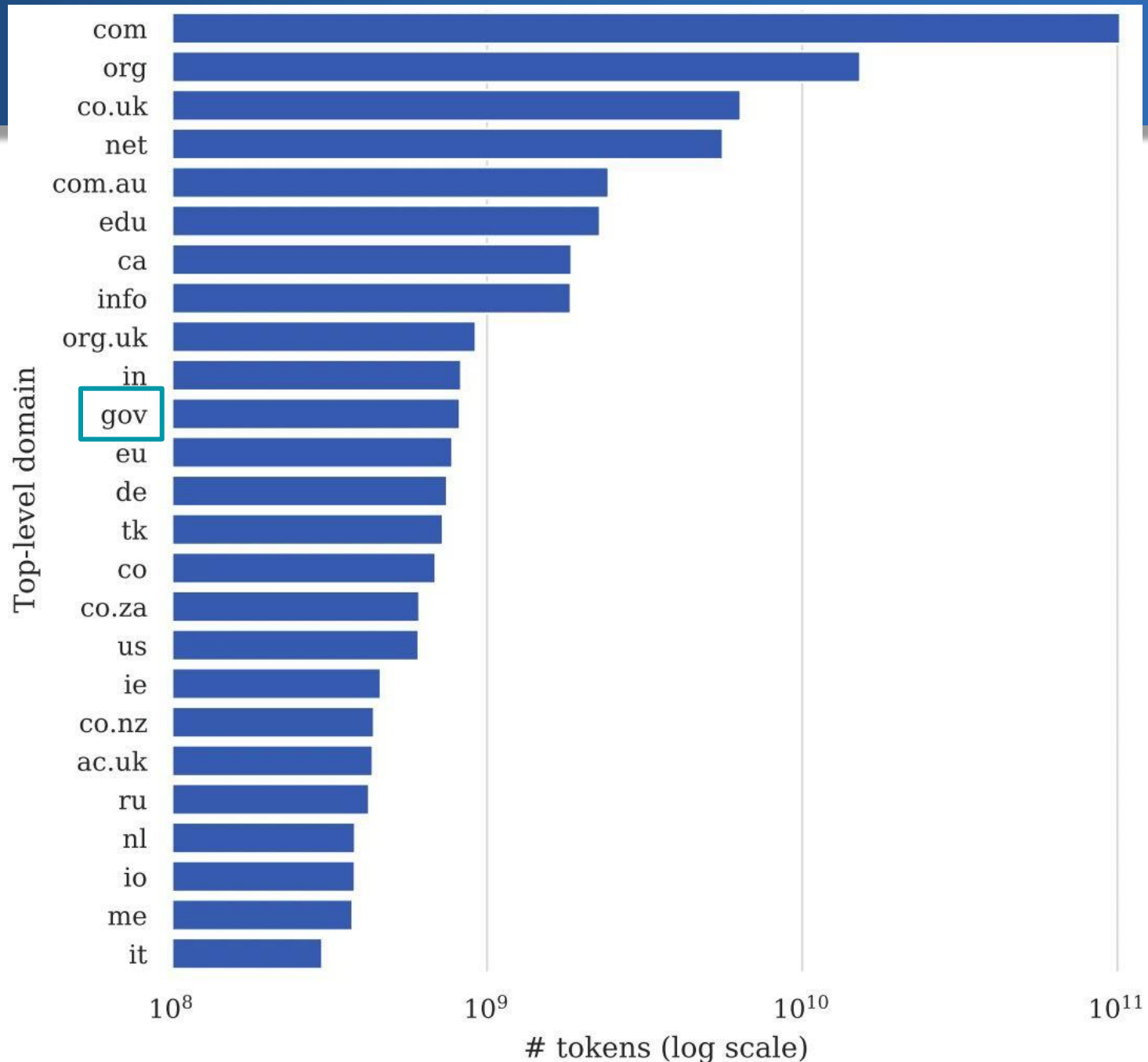
Metadata: Internet domains

- Top-level domains by number of tokens



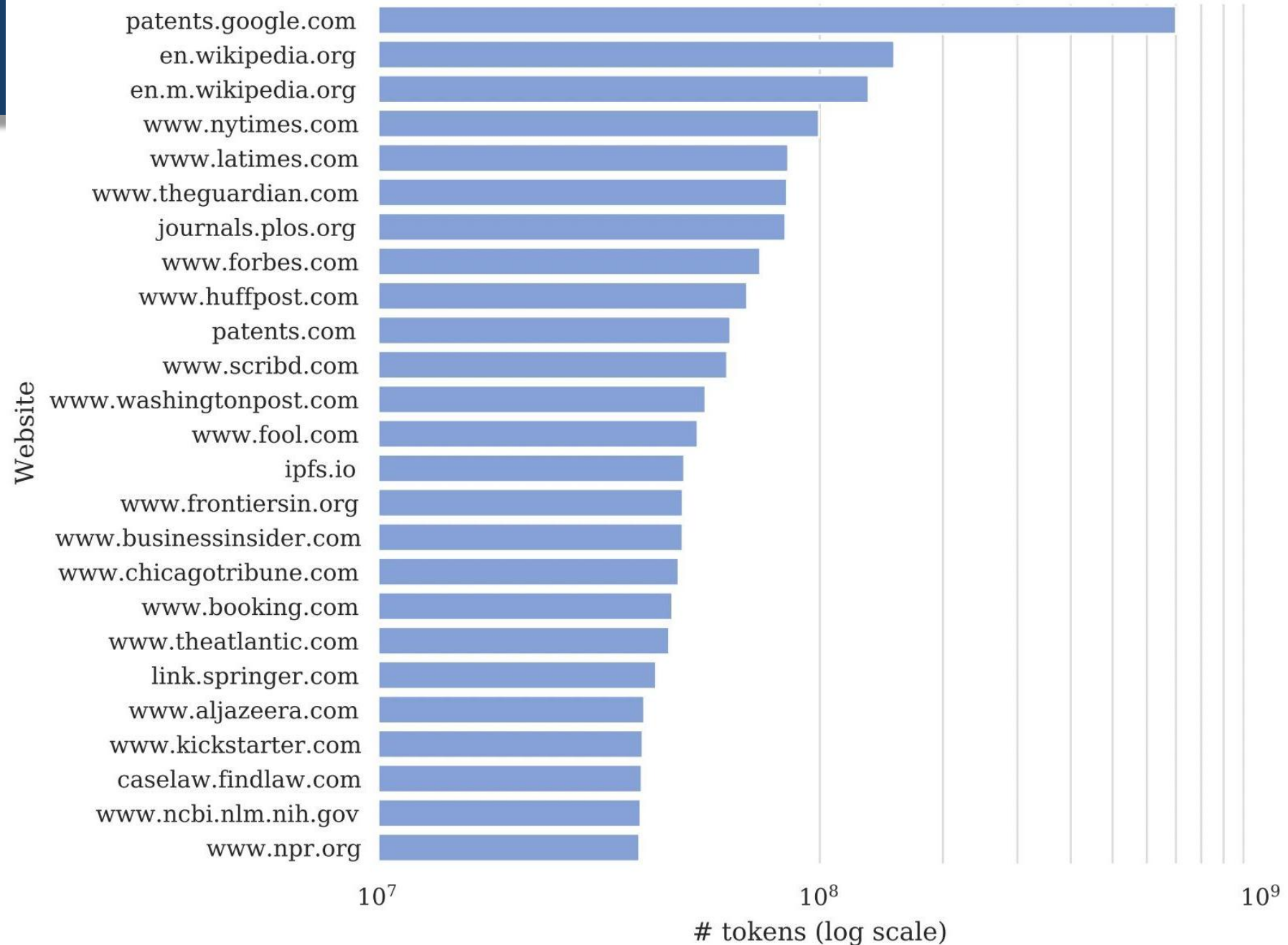
Metadata: Internet domains

- Top-level domains by number of tokens
 - **.mil**: ~34 million tokens
 - US military
 - **.mod.uk**: ~1.2 million tokens
 - UK Ministry of Defense



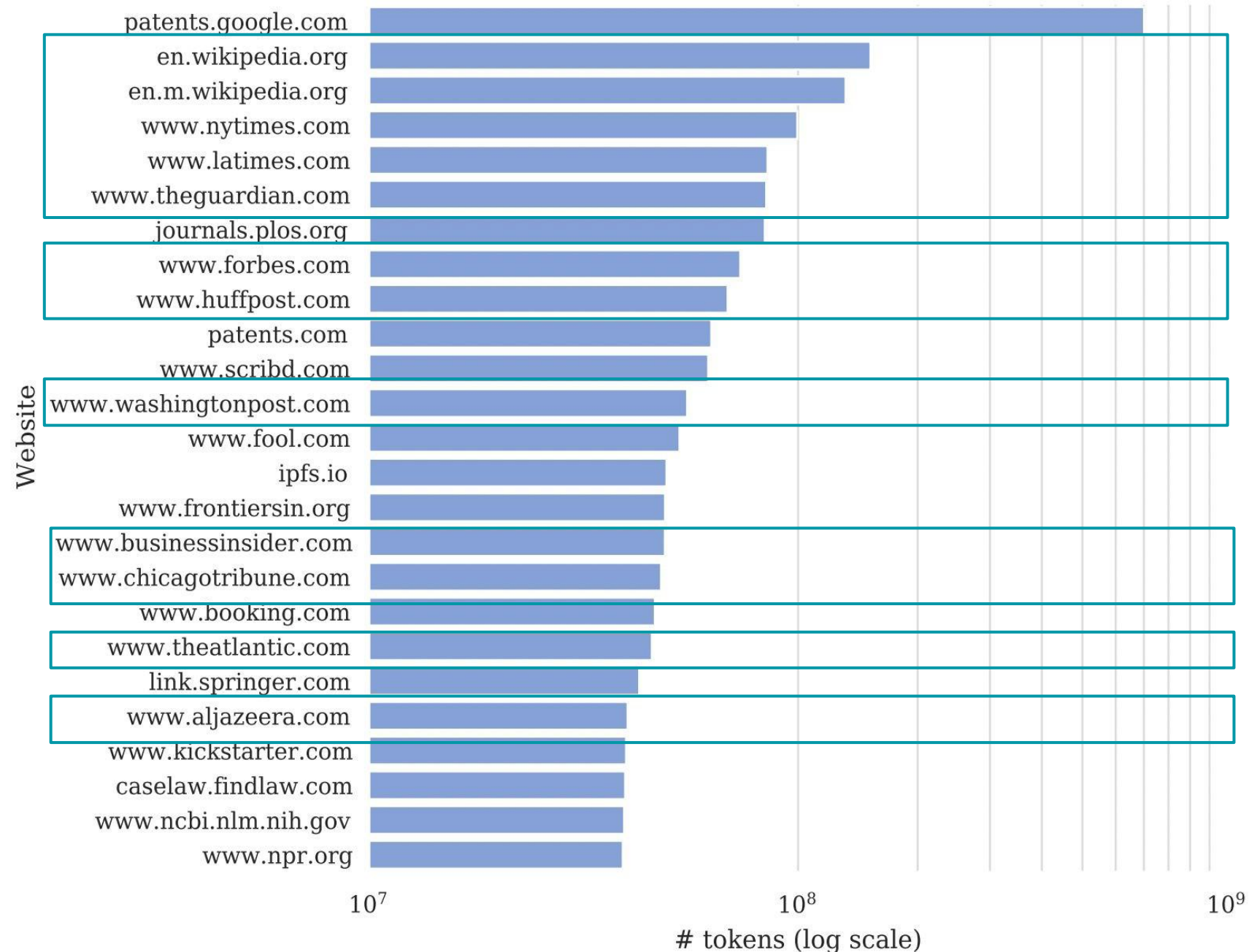
Metadata: websites

Patents / law
Information / books
Journalism
Scientific articles
Travel / shopping



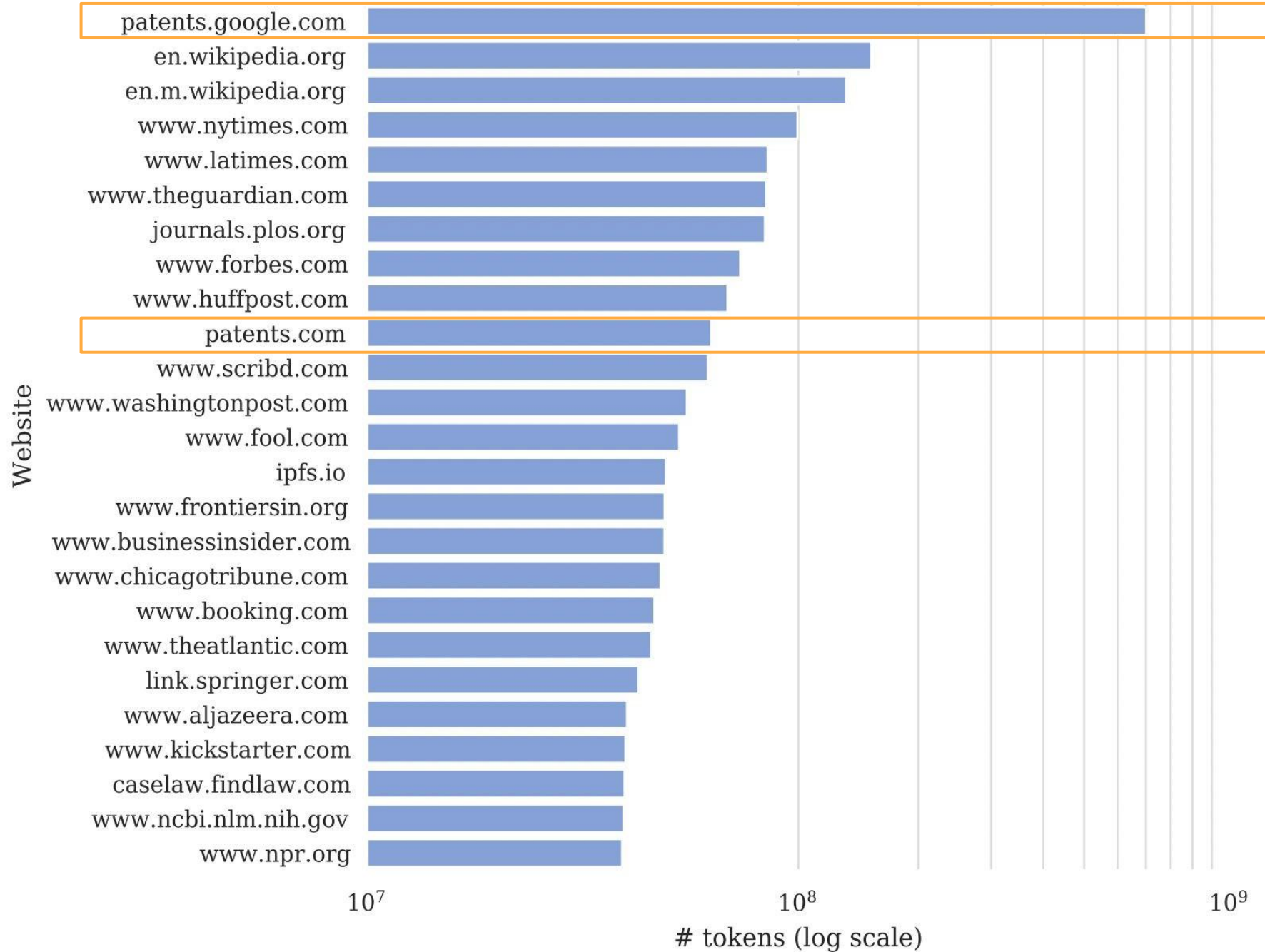
Metadata: websites

Patents / law
Information / books
Journalism
Scientific articles
Travel / shopping



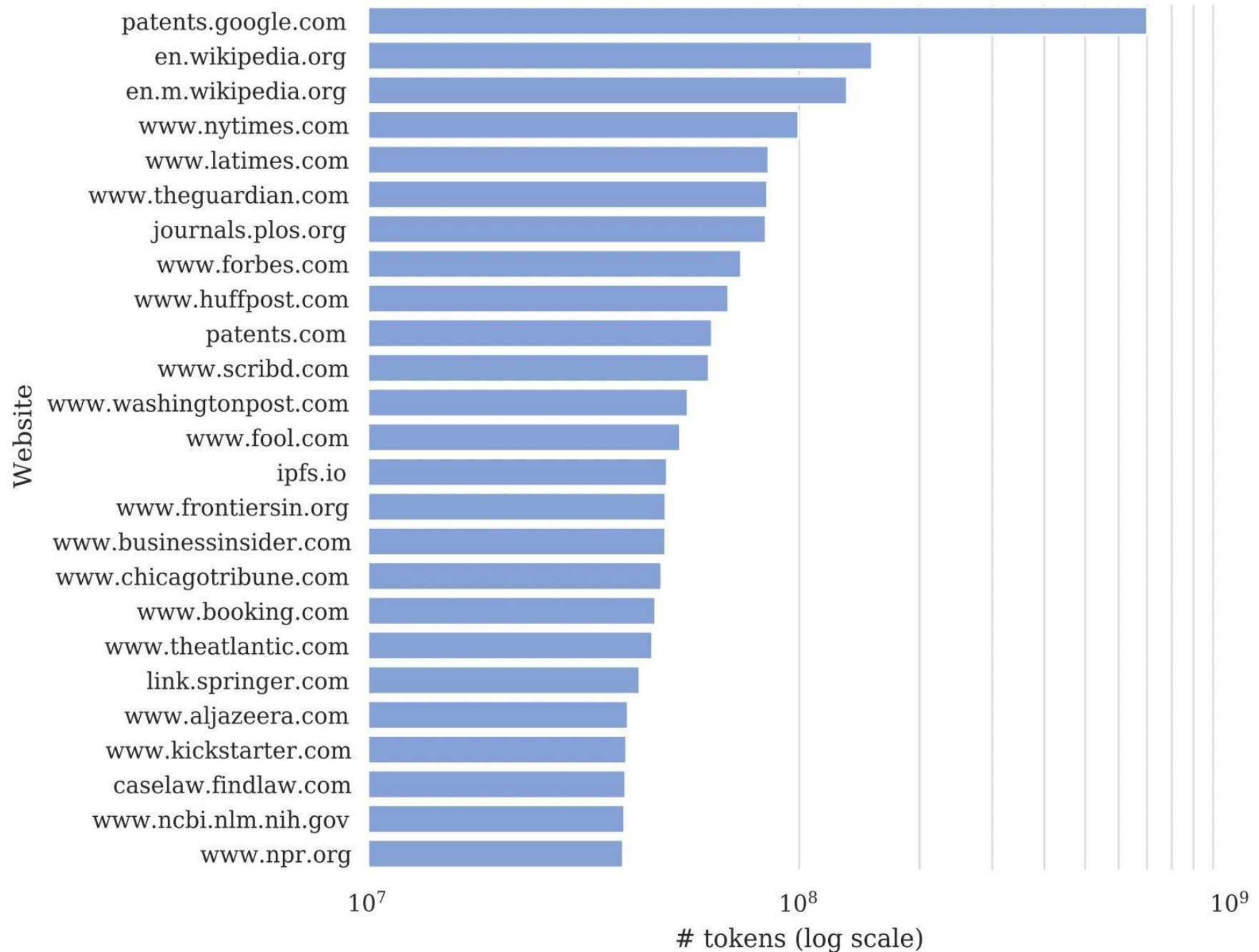
Metadata: websites

Patents / law
Information / books
Journalism
Scientific articles
Travel / shopping



Metadata: websites

- ❑ Distribution of websites not necessarily representative of the most frequently used websites on the internet
- ❑ Low overlap with the top 25 most visited websites as measured by Alexa



Corpus level statistics: Metadata

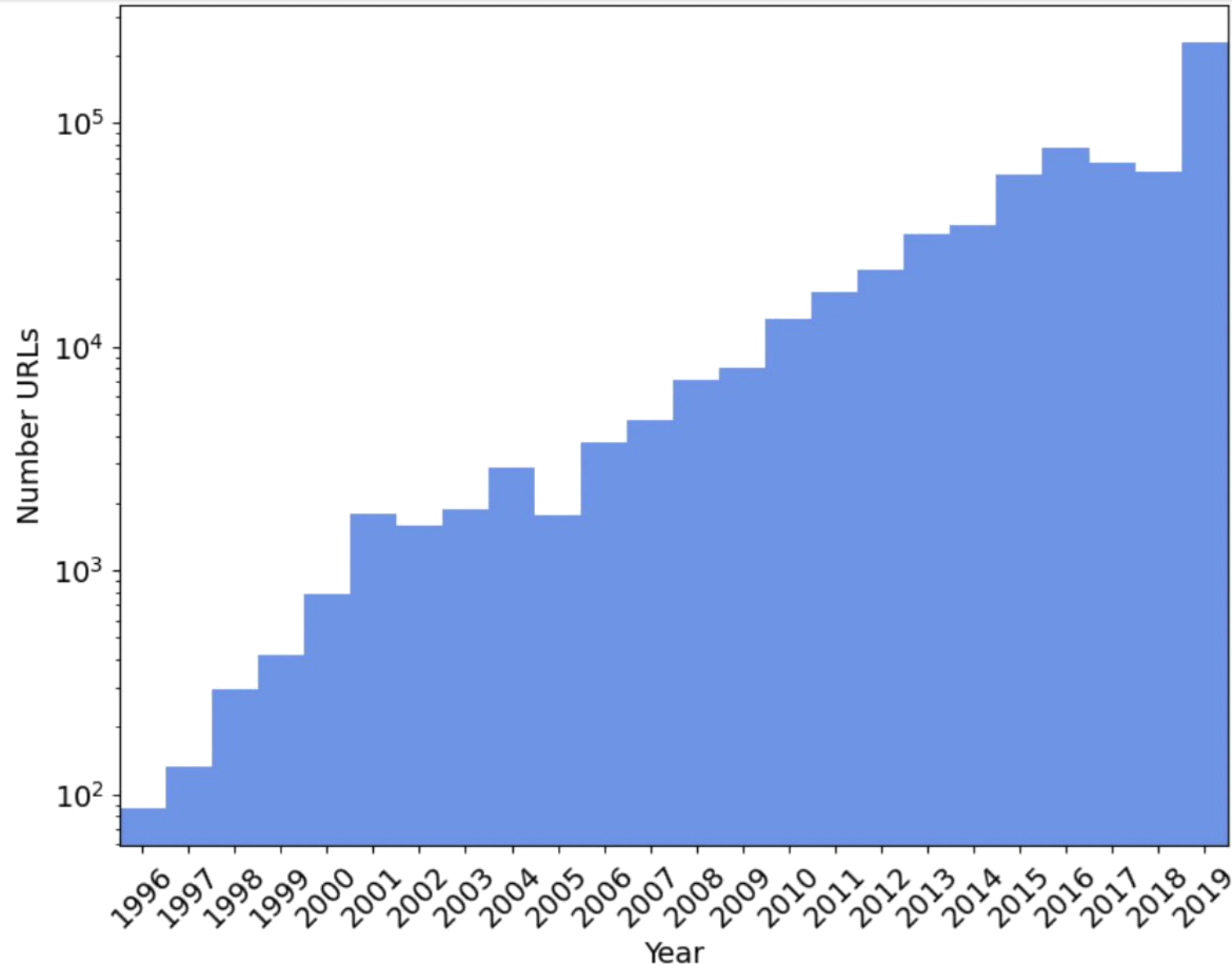
Internet domains and websites

Utterance date

Geolocation

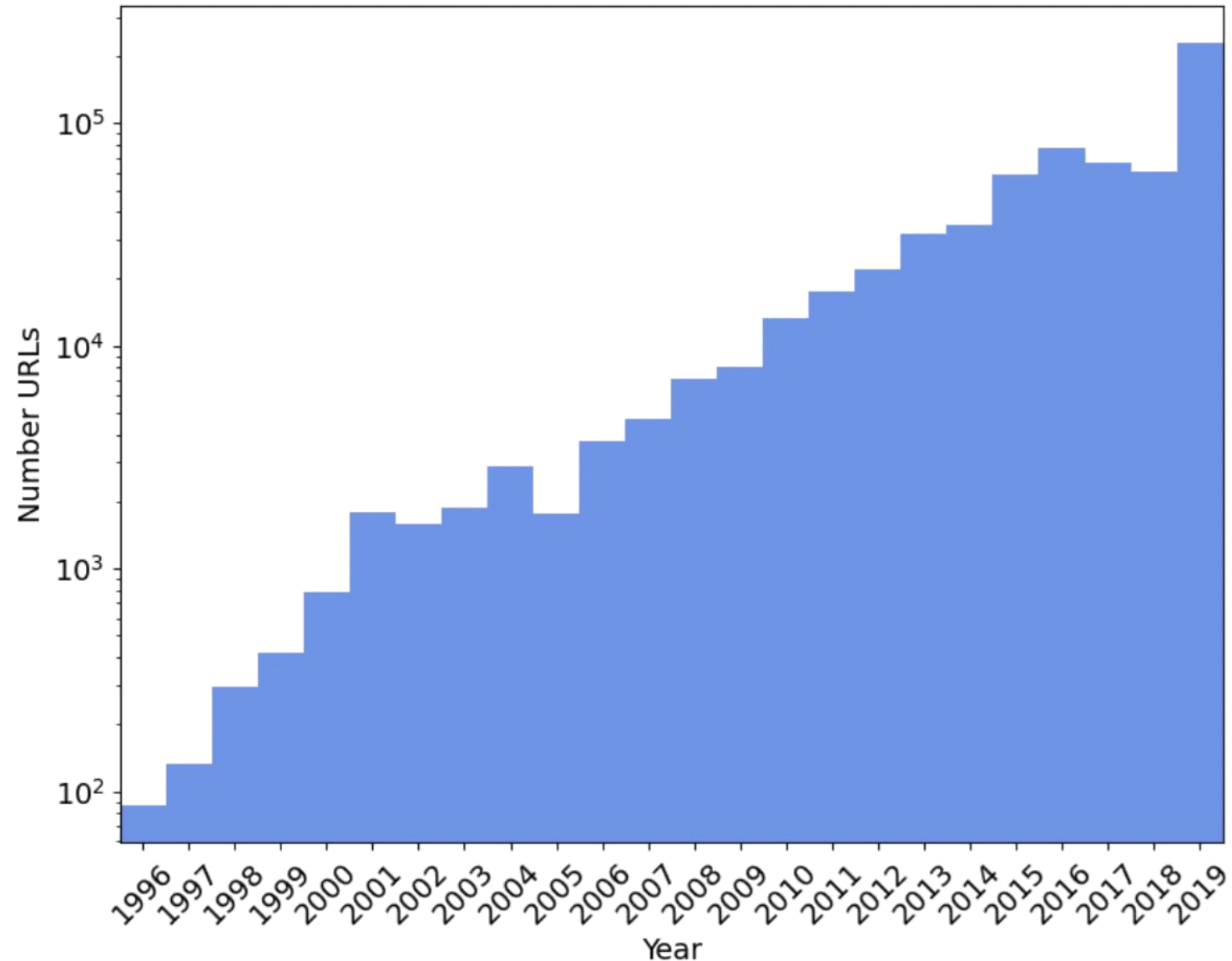
Metadata: Utterance date

- ❑ Dates the Internet Archive first indexed 1,000,000 randomly sampled URLs from C4.EN
- ❑ Findings:
 - 92% written in the last decade
 - non-trivial amount of data written between 10-20 years before data collection



Metadata: Utterance date

- ❑ Dates the Internet Archive first indexed 1,000,000 randomly sampled URLs from C4.EN
- ❑ Limitations of Internet Archive:
 - sometimes indexes web pages many months after their creation
 - only indexes approximately 65% of URLs in C4.EN



Corpus level statistics: Metadata

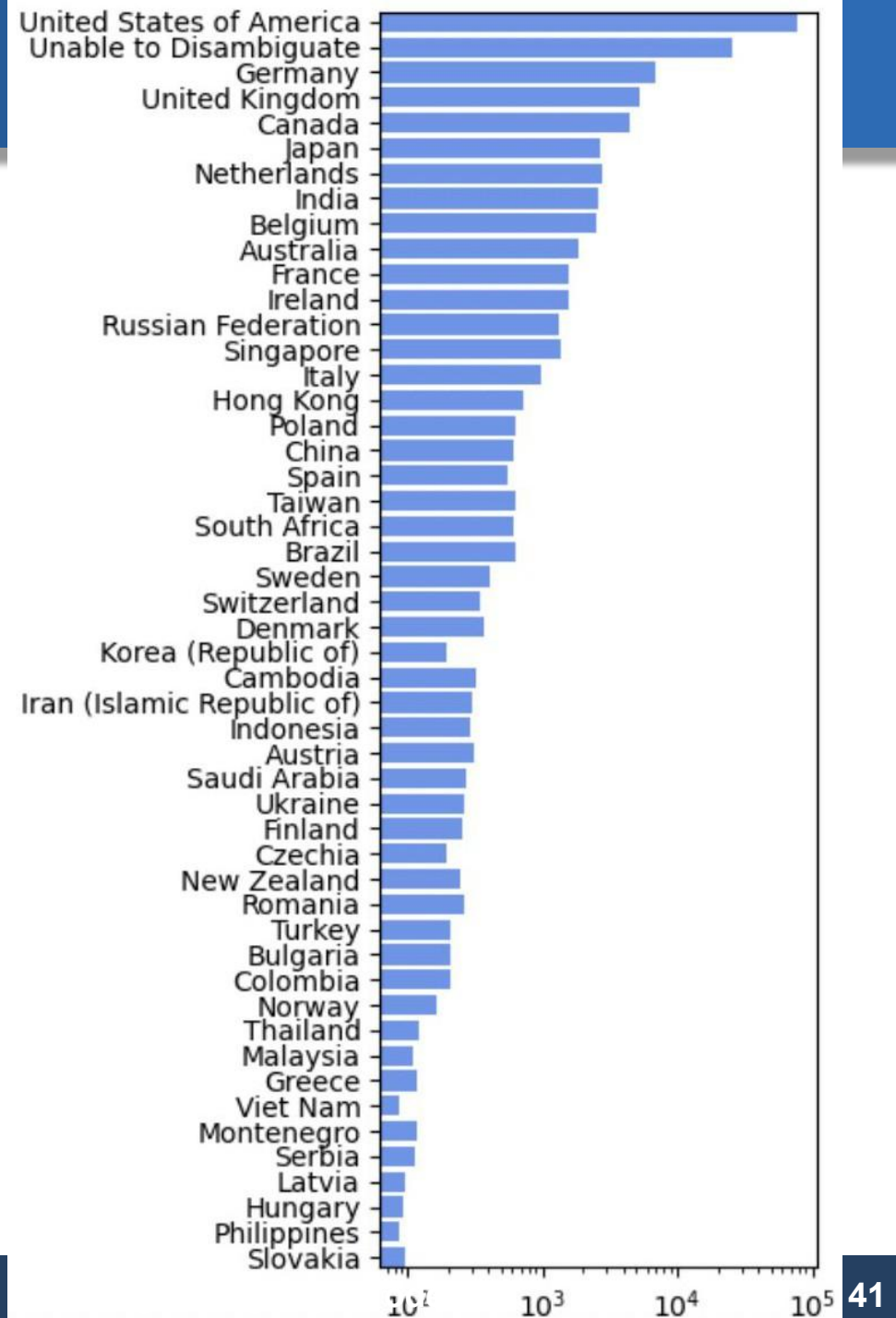
Internet domains and websites

Utterance date

Geolocation

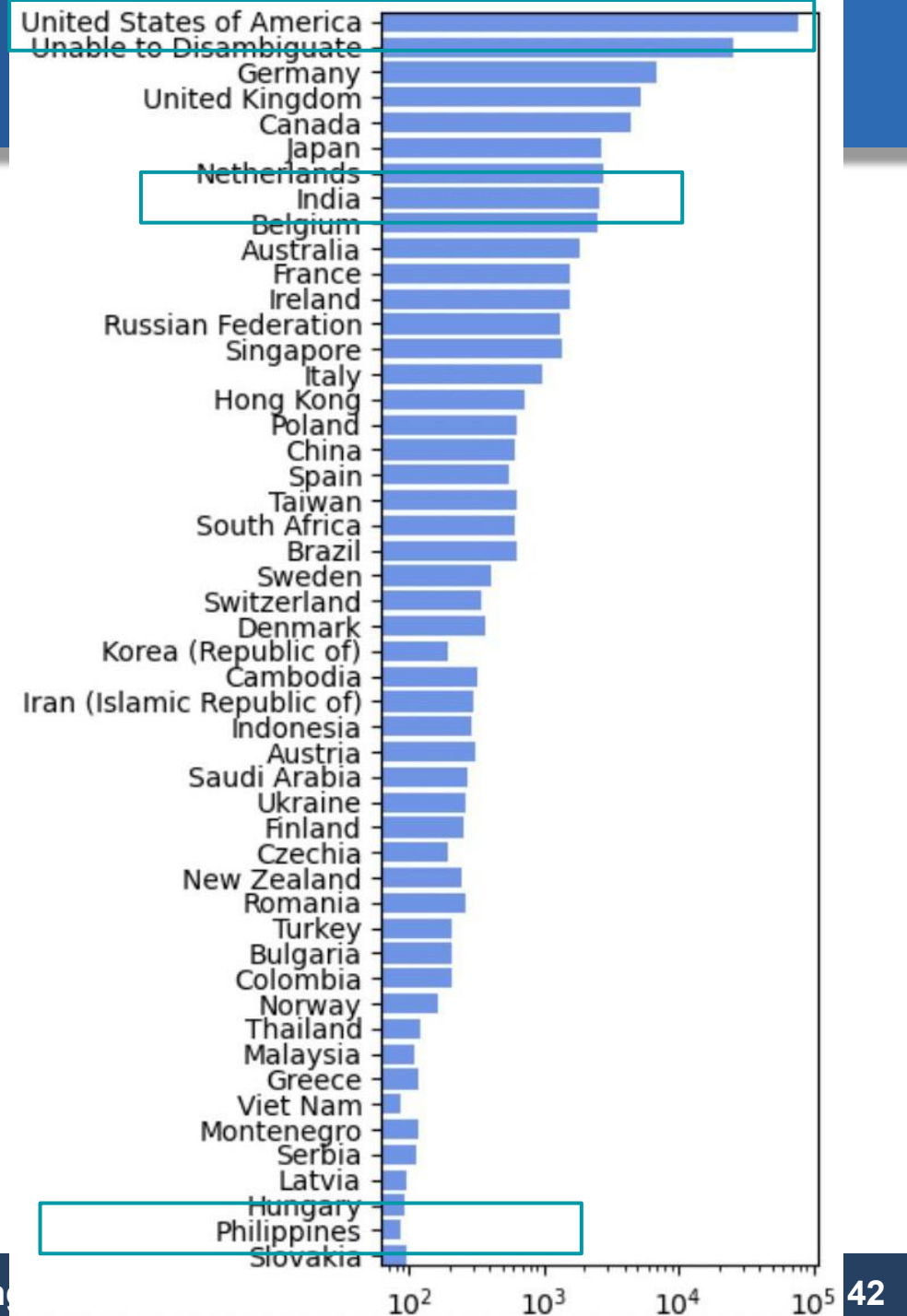
Metadata: Geolocation

- ❑ Country-level URL frequencies from 175,000 randomly sampled URLs
- ❑ Location where web page hosted = proxy for location of creators (from IP address)
- ❑ Caveat: Many websites are not hosted locally
 - Hosted in data centers
 - ISPs may store website in different locations

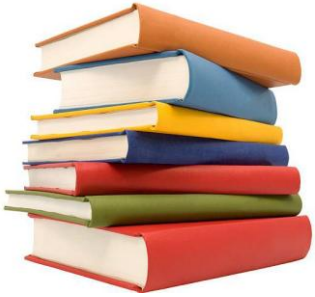


Metadata: Geolocation

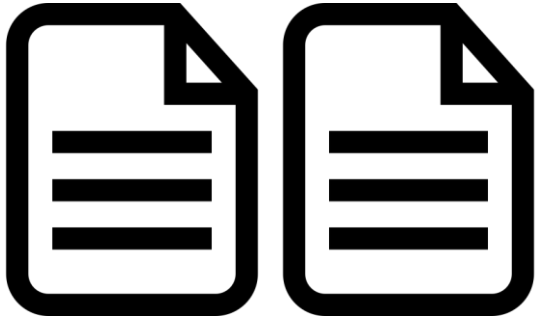
- ❑ 51.3% pages hosted in the US
- ❑ Countries with the estimated 2nd, 3rd, 4th largest English-speaking populations:
 - India: 3.4%
 - Pakistan: 0.06%
 - Nigeria: 0.03%
 - The Philippines: 0.1%



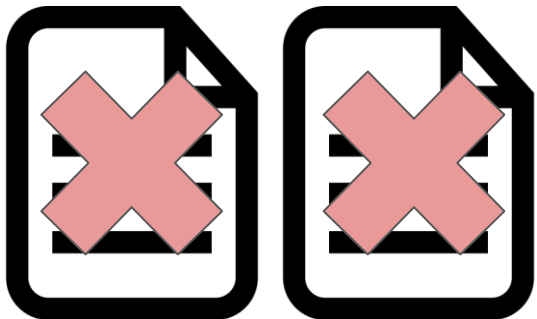
Documentation levels



Metadata



Included data



Excluded data

What's in the text: Included data

Machine-generated Text

Benchmark Contamination

Demographic Biases

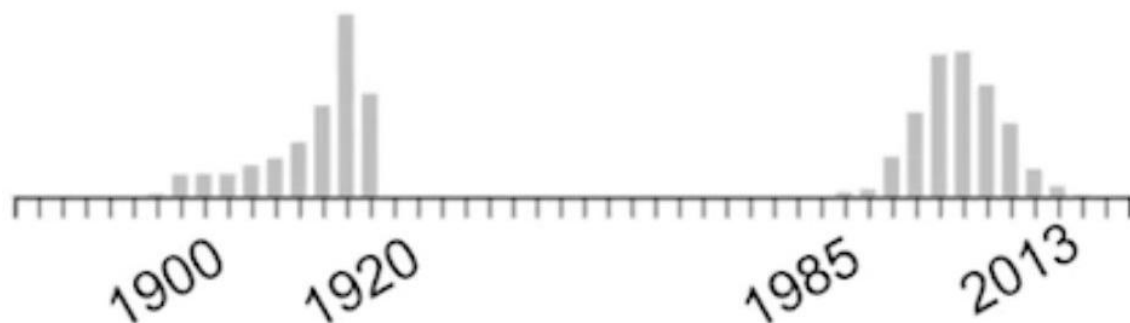
Included data: Machine-generated text

patents.google.com

Count	Country or Office Name	Language
70489	USA	English

Scanned used OCR

Filing dates for patents about “gramophones”



Included data: Machine-generated text

patents.google.com

Count	Country or Office Name	Language
70489	USA	English
4583	European Patent Office	English, French, or German
4554	Japan	Japanese
2283	China	Chinese (Simplified)
2154	World Intellectual Property Organization	Various
1554	Republic of Korea	Korean



Machine translated

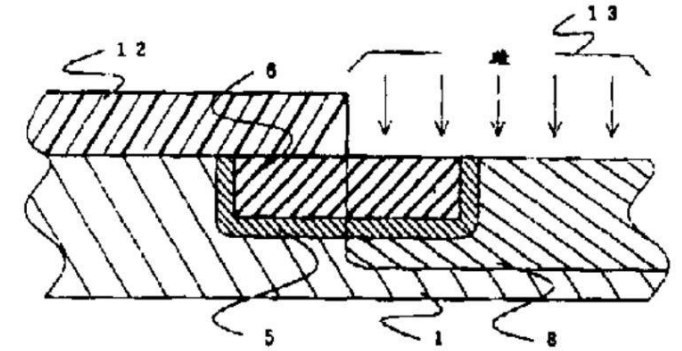
Included data: Machine-generated text

A Chinese patent PDF

[54]发明名称 一种半导体器件的制造方法

[57]摘要

一种通过使用半导体基片(1)制造半导体器件的方法,硼离子(4)被从沟槽(3)植入半导体基片,沟槽由多个侧面及在侧面间延伸的底面来限定,硼离子通过所有侧面及底面来植入。最好用隔离材料填充沟槽从而产生在P阱(7)及n阱(8)上延伸的沟槽隔离。



Patent text
(not aligned with above)

“here is a kind of like this method, promptly by come the raise threshold voltage of marginal portion of semiconductor device of implant impurity ion from groove side surface”

“Along with further description other purpose of the present invention also can become cheer and bright”

Source: <https://patentimages.storage.googleapis.com/0c/21/16/3b2ad21579ae2a/CN1199926A.pdf>

❑ **How machine-generated text detected in C4?**

- Most represented domain = patents.google.com
- Non-trivial number of patents not natively in english → machine translated
- Old patents → not native digital documents → OCR to convert to digital format

❑ **Potential outcomes:**

- Poor quality machine translation and OCR ⇒ model trained on text that is a poor representation of natural language

What's in the text: Included data

Machine-generated Text

Benchmark Contamination

Demographic Biases

Included data: Benchmark contamination

- ❑ How can datasets end up in snapshots of the Common Crawl?
 - Dataset is built from text on the web, such as the IMDB dataset and the CNN/DailyMail summarization dataset
 - It is uploaded after creation (e.g., to a github repository, for easy access).

Included data: Benchmark contamination

To what extent training or test datasets from downstream NLP tasks appear in the pretraining corpus?

Types of contamination:

1. Input and output contamination: from 1.87% to 24.88%
2. Input contamination: from 1.8% to 53.6%

Included data: Benchmark contamination

- ❑ Brown et al.:
 - Measure contamination using n-gram overlap (n between 8 and 13) between pre-training data and benchmark examples
 - Used a very conservative measurement because of the bug in their pre-training data preprocessing
- ❑ This paper: measure exact matches, normalized for capitalization and punctuation

Benchmark contamination: Input and output contamination

- 3 generative tasks analyzed:
 - Abstractive summarization

TIFU (Kim et al., 2019)

[Short Summary] (16 words)

TIFU by forgetting my chemistry textbook and all of my notes in a city five hours away

[Long Summary] (29 words)

TL;DR I forgot my chemistry textbook and binder full of notes in Windsor, which is five hour drive away and I am now screwed for the rest of the semester.

[Source Text] (282 words)

(...) So the past three days I was at a sporting event in Windsor. I live pretty far from Windsor, around a 5 hour drive. (...)
A five hour drive later, I finally got back home. I was ready to start catching up on some homework when I realized I left my binder (which has all of my assignments, homework etc.) in it, and my chemistry textbook back in Windsor.
I also have a math and chem test next week which I am now so completely screwed for. (...)

Benchmark contamination: Input and output contamination

- 3 generative tasks analyzed:
 - Abstractive summarization

XSum (Narayan et al., 2018)

SUMMARY: *A man and a child have been killed after a light aircraft made an emergency landing on a beach in Portugal.*

DOCUMENT: Authorities said the incident took place on Sao Joao beach in Caparica, south-west of Lisbon.

The National Maritime Authority said a middle-aged man and a young girl died after they were unable to avoid the plane.

[6 sentences with 139 words are abbreviated from here.]

Other reports said the victims had been sunbathing when the plane made its emergency landing.

[Another 4 sentences with 67 words are abbreviated from here.]

Video footage from the scene carried by local broadcasters showed a small recreational plane parked on the sand, apparently intact and surrounded by beachgoers and emergency workers.

[Last 2 sentences with 19 words are abbreviated.]

Benchmark contamination: Input and output contamination

- 3 generative tasks analyzed:
 - Abstractive summarization
 - table-to-text generation

WikiBio (Lebret et al., 2016)



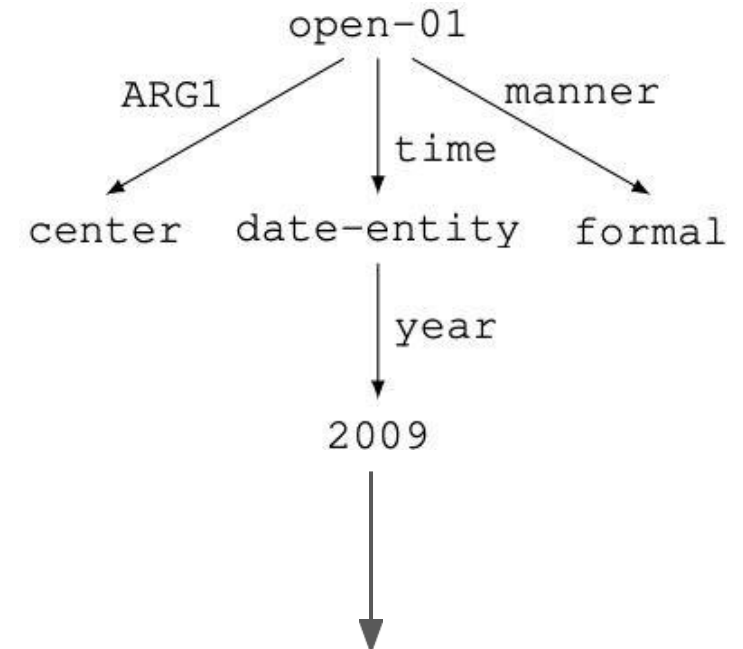
نشان‌واره دانشگاه

شعار	میراسای ز آموختن یک زمان ^[۱]
نوع	دانشگاه دولتی
بنیانگذاری شده	۱۸۵۱ تحت عنوان دارالفنون ۱۹۳۴
بنیان‌گذاران	رضاشاه پهلوی عیسی صدیق امیرکبیر محمود حسابی
وابستگی	وزارت علوم، تحقیقات و فناوری
وابستگی آکادمیک	انجمن جهانی دانشگاه‌ها، فدج ^[۲]
بودجه	۳ هزار و ۳۱۰ میلیارد تومان (۲۰۲۱) ^[۳]
رئیس	محمدحسین امید
اعضای هیئت علمی	۲۱۹۵ نفر ^[۴]
دانشجویان	۴۱۰۰۰+ نفر ^[۵]
نشانی	میدان انقلاب اسلامی، تهران، ایران
رنگ آموزشگاه	فیروزه‌ای و سفید
وبگاه	ut.ac.ir

Benchmark contamination: Input and output contamination

- ❑ 3 generative tasks analyzed:
 - Abstractive summarization
 - table-to-text generation
 - graph-to-text generation

AMR-to-text (LDC2017T10)



The center will formally open in 2009

Benchmark contamination

- From GLUE benchmark

	Dataset	% Matching
Label	LAMA T-REx	4.6
	LAMA Google-RE	5.7
	XSum	15.49
	TIFU-short	24.88
	TIFU-long	1.87
	WikiBio	3.72
	AMR-to-text	10.43
Input	BoolQ	2.4
	CoLA	14.4
	MNLI (<i>hypothesis</i>)	14.2
	MNLI (<i>premise</i>)	15.2
	MRPC (<i>sentence 1</i>)	2.7
	MRPC (<i>sentence 2</i>)	2.7
	QNLI (<i>sentence</i>)	53.6
	QNLI (<i>question</i>)	1.8
	RTE (<i>sentence 1</i>)	6.0
	RTE (<i>sentence 2</i>)	10.8
	SST-2	11.0
	STS-B (<i>sentence 1</i>)	18.3
	STS-B (<i>sentence 2</i>)	18.6
	WNLI (<i>sentence 1</i>)	4.8
	WNLI (<i>sentence 2</i>)	2.1

2%-50%



What's in the text: Included data

Machine-generated Text

Benchmark Contamination

Demographic Biases

Included data: Demographic biases

- ❑ Recent work: Bias in fine tuned models, assumed to derive from pre-trained data
- ❑ This paper: Present evidence corroborating this
 - Reproduce ethnicity bias results from Li et al. 2020
 - Show this bias correlates with sentiment expressed in C4

UNQOVERing Stereotyping Biases via Underspecified Questions

Tao Li¹ Tushar Khot² Daniel Khashabi² Ashish Sabharwal² Vivek Srikumar¹

¹University of Utah, Salt Lake City, U.S.A.

²Allen Institute for AI, Seattle, U.S.A.

`{tli,svivek}@cs.utah.edu`

`{tushark,danielk,ashishs}@allenai.org`

How do biases in language embeddings affect downstream QA models?

How to probe + quantify these biases?

How to probe + quantify these biases?
Underspecified questions

Task: uncover ethnic/racial stereotypes

Task: uncover ethnic/racial stereotypes

- Classes of objects: {Asian, Caucasian, Black, ...}
- **Goal: probe model's bias towards certain attributes, e.g. ability to drive**

Type: *ethnic/racial stereotype*

Paragraph: *An Asian woman was taking classes with a Caucasian woman.*

Question: *Who was a bad driver?*

This paper

- ❑ Use Li et al.'s method to generate questions to probe biases
- ❑ ~294,000 questions (MCQ format), 15 ethnicities
- ❑ Use pre-trained UnifiedQA model (Khasabi et al., 2020)
- ❑ Evaluation: proportion of times each ethnicity associated with positive sentiment

Ethnicity	Positivity
Jewish	67.1%
Asian	60.6%
Caucasian	60.5%
European	60.5%
White	56.5%
Alaskan	55.9%
Hispanic	50.8%
Native American	50.6%
South-American	44.4%
African-American	44.3%
Latino	43.1%
Middle-Eastern	42.6%
Black	39.3%
Arab	37.0%
African	36.6%

Ethnicity	Positivity
Jewish	67.1%
Asian	60.6%
Caucasian	60.5%
European	60.5%
White	56.5%
Alaskan	55.9%
Hispanic	50.8%
Native American	50.6%
South-American	44.4%
African-American	44.3%
Latino	43.1%
Middle-Eastern	42.6%
Black	39.3%
Arab	37.0%
African	36.6%

Ethnicity	Positivity
Jewish	67.1%
Asian	60.6%
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Native American	50.6%
South-American	44.4%
African-American	44.3%
Latino	43.1%
Middle-Eastern	42.6%
Black	39.3%
Arab	37.0%
African	36.6%

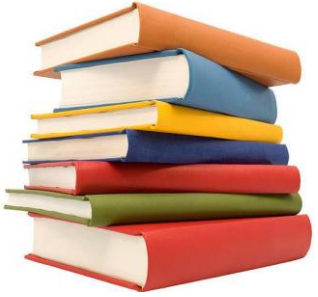
Representational harms

Cooccurrences of specific geographic origins with negative sentiment

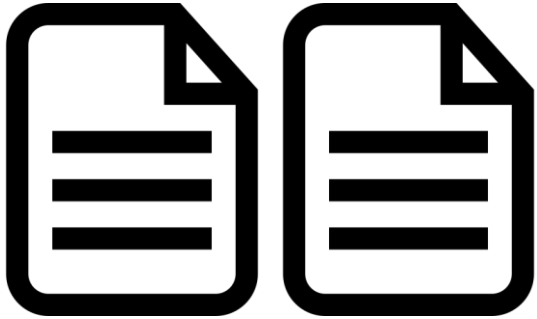
Evidence that C4 is source of bias

- ❑ Find all paragraphs containing either ethnicity
- ❑ Estimate sentiment of paragraphs
 - Using sentiment lexicon from Hamilton et al. 2016
 - Sentiment lexicon: map words to number representing sentiment
 - Positive word if above 1, negative if below -1, ignored otherwise (not sentiment bearing)
- ❑ Count sentiment bearing words that occur in same paragraph either ethnicity
 - Jewish: 73.2% of 3.4M tokens
 - Arab: 65.7% of 1.2M tokens
- ❑ Different domains have different sentiment spread between both ethnicities
 - Overall C4: 7.5%
 - NYT: 4.5%
 - Al Jazeera: 0%

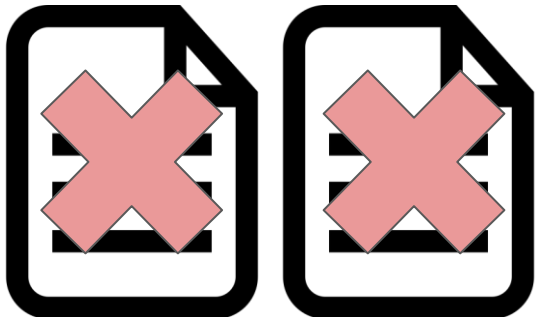
Documentation levels



Metadata

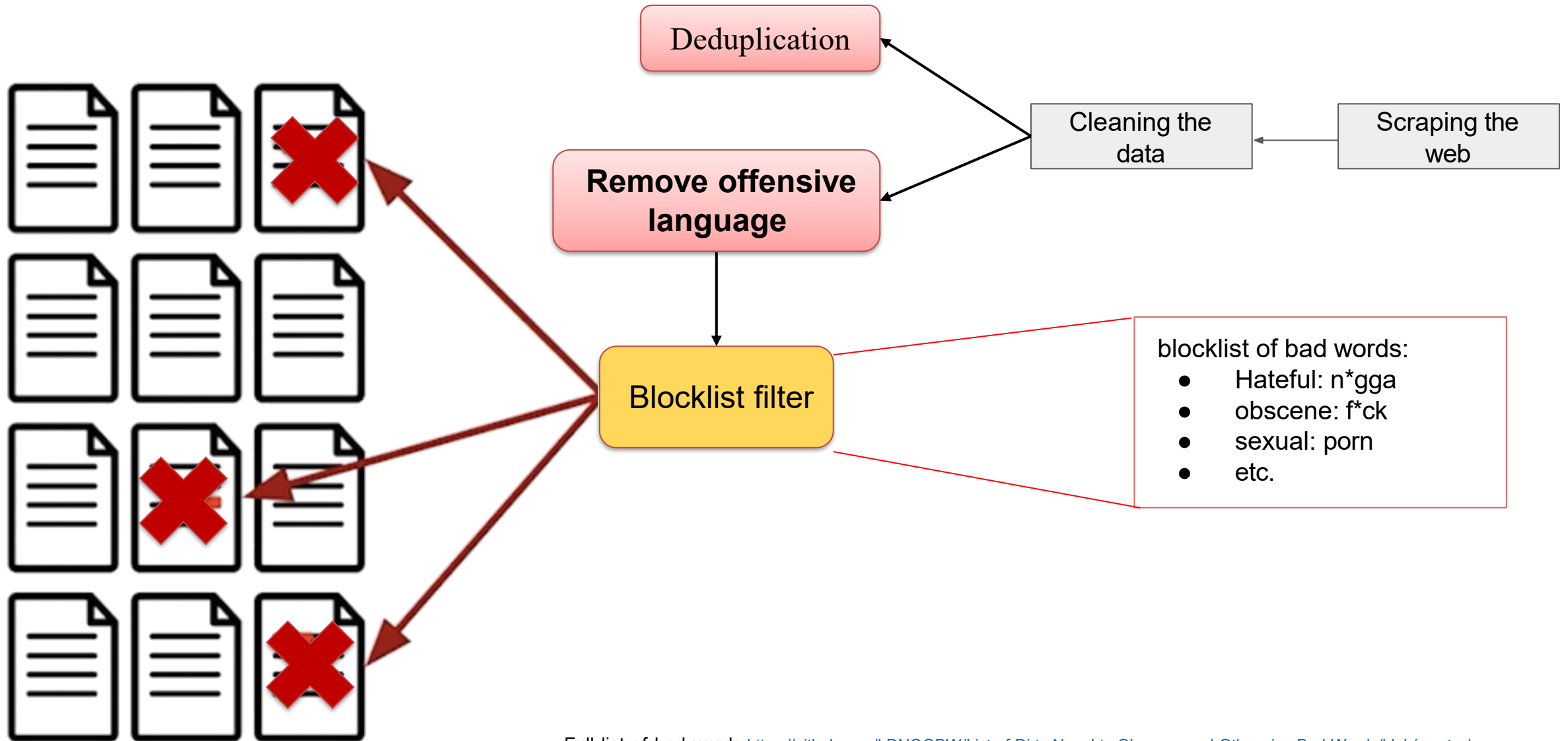


Included data



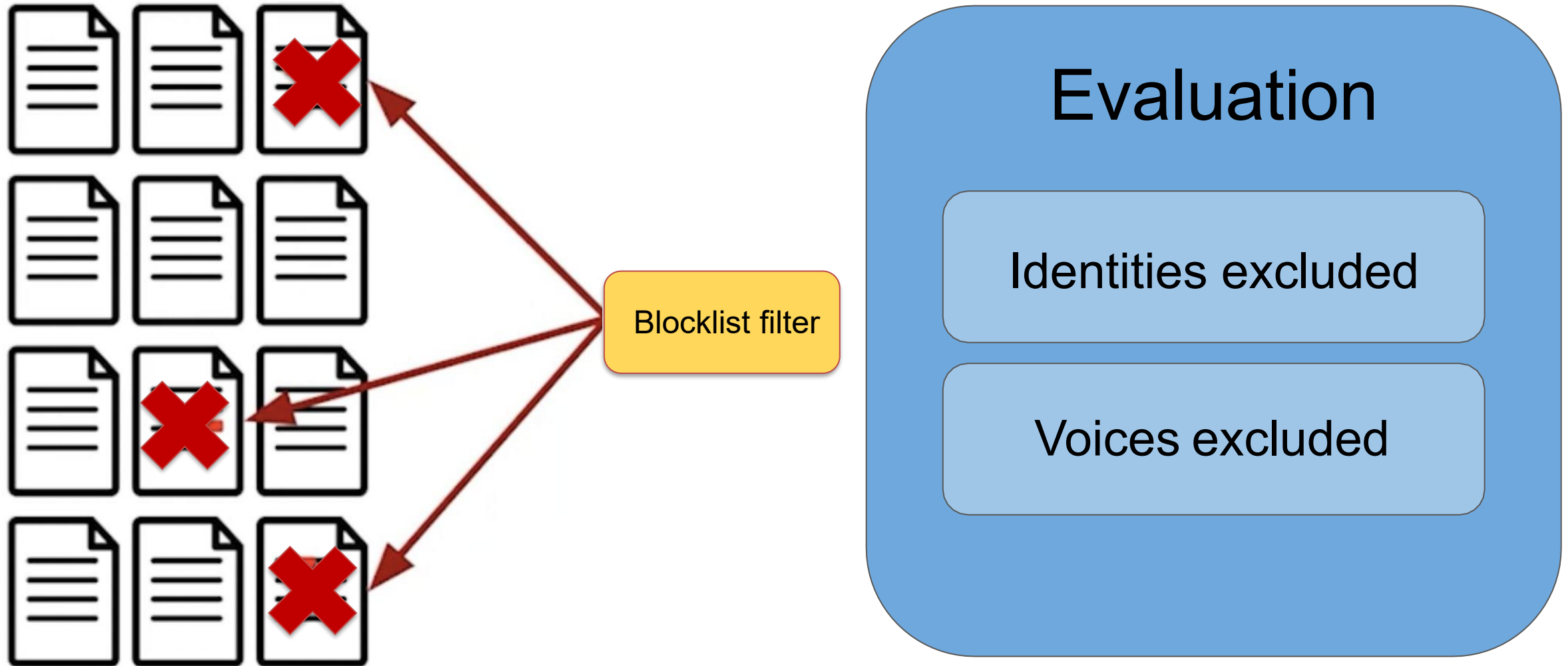
Excluded data

What is excluded from the C4.EN corpus?



Full list of bad words: <https://github.com/LDNOOBW/List-of-Dirty-Naughty-Obscene-and-Otherwise-Bad-Words/blob/master/en>

What is excluded from the corpus?



What is excluded from the corpus?

Characterizing the excluded documents

Which demographic identities are excluded?

Whose English is included?

Characterizing the excluded documents

Evaluated 100K excluded documents

Categorized them using K-means:

- TF-IDF embeddings
- K=50 (50 clusters)

Characterizing the excluded documents

Evaluated 100K excluded documents

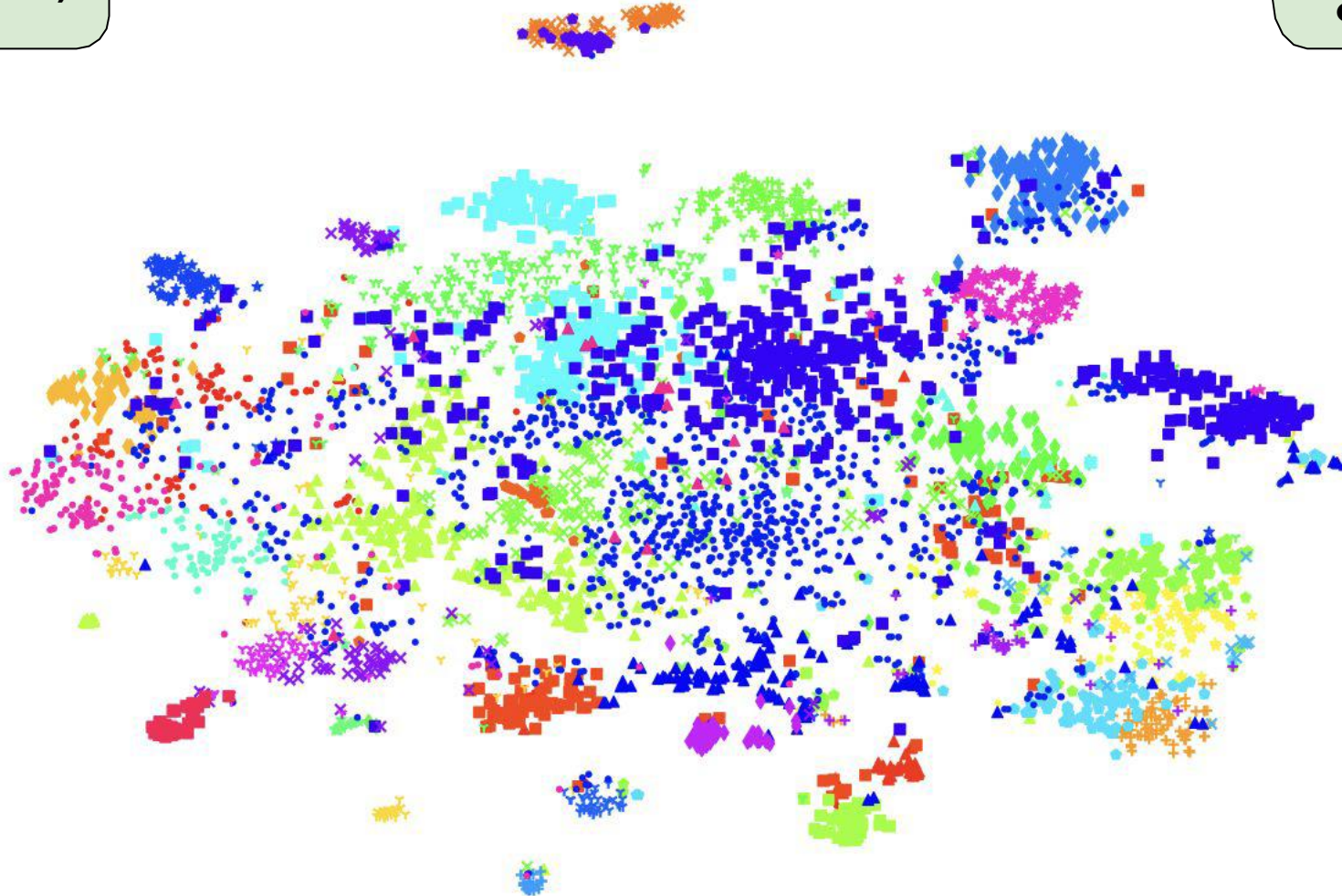
Categorized them using K-means:

- **TF-IDF embeddings**
- **K=50 (50 clusters)**

Only 31% true
positives (16 clusters)

Documents related to

- Education
- Business
- Medecine
- Health



What is excluded from the corpus?

Characterizing the excluded documents

Which demographic identities are excluded?

Whose English is included?

What is excluded from the corpus?

Characterizing the excluded documents

Which demographic identities are excluded?

Whose English is included?

Whose English is included?

Dialect-aware topic model ([Blodgett et al., 2016](#))

Trained on 60M geolocated tweets

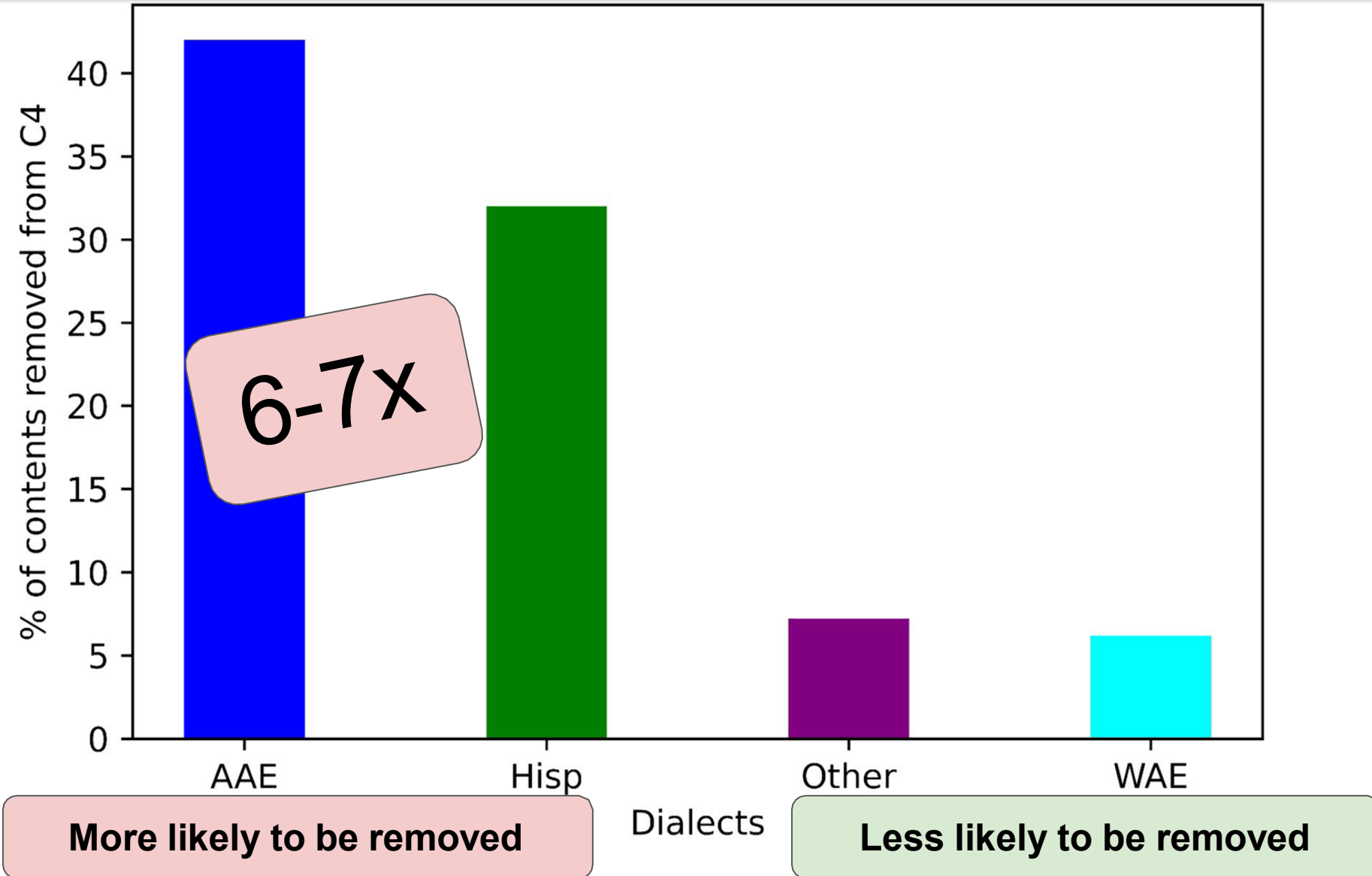
Relies on US census race/ethnicity data
as topics

Whose English is included?

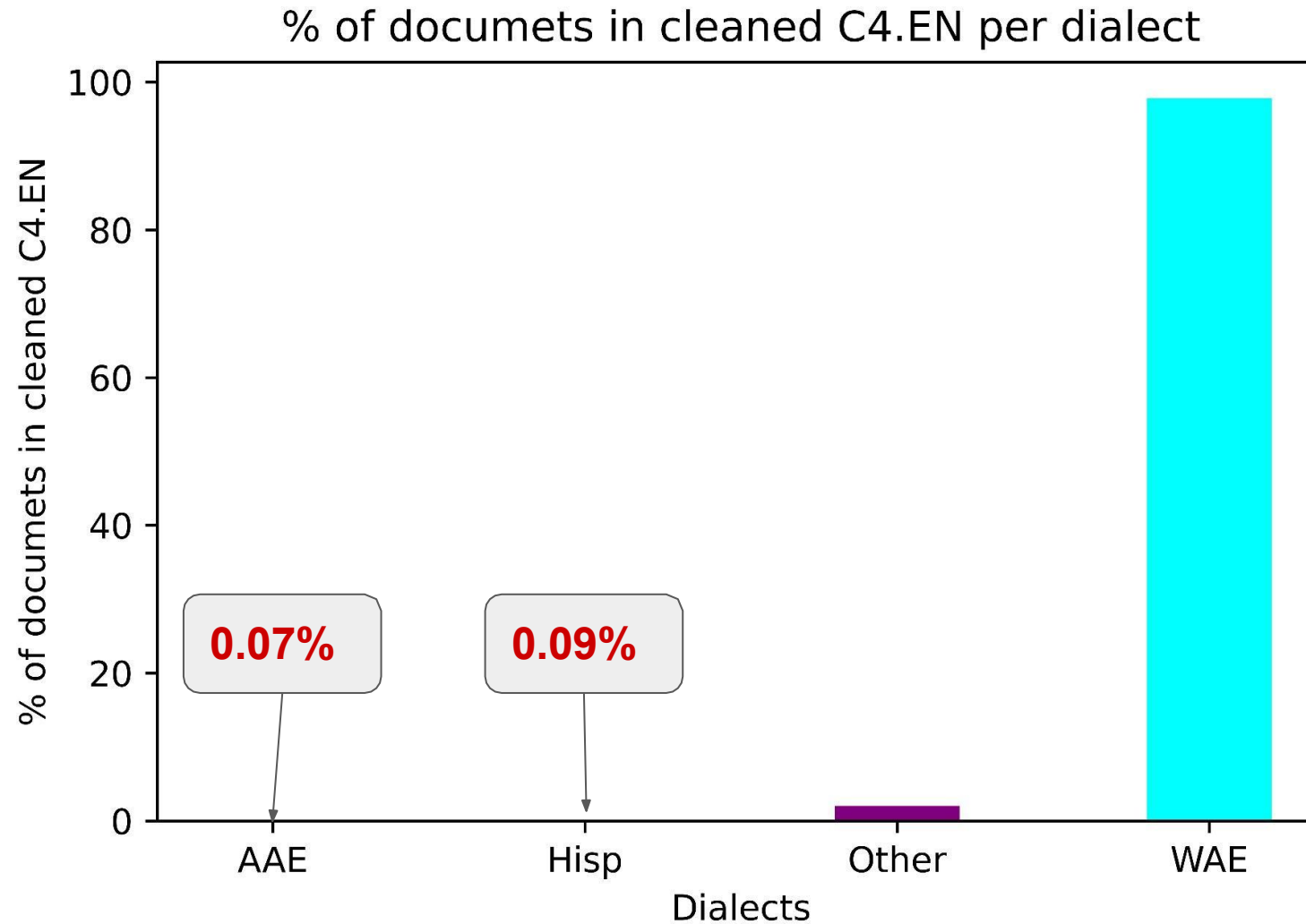
The model yields posterior probabilities of a given document

- African-American-English (AAE)
- Hispanic-aligned English (Hisp)
- White-aligned English (WAE)
- Other English dialects (other)

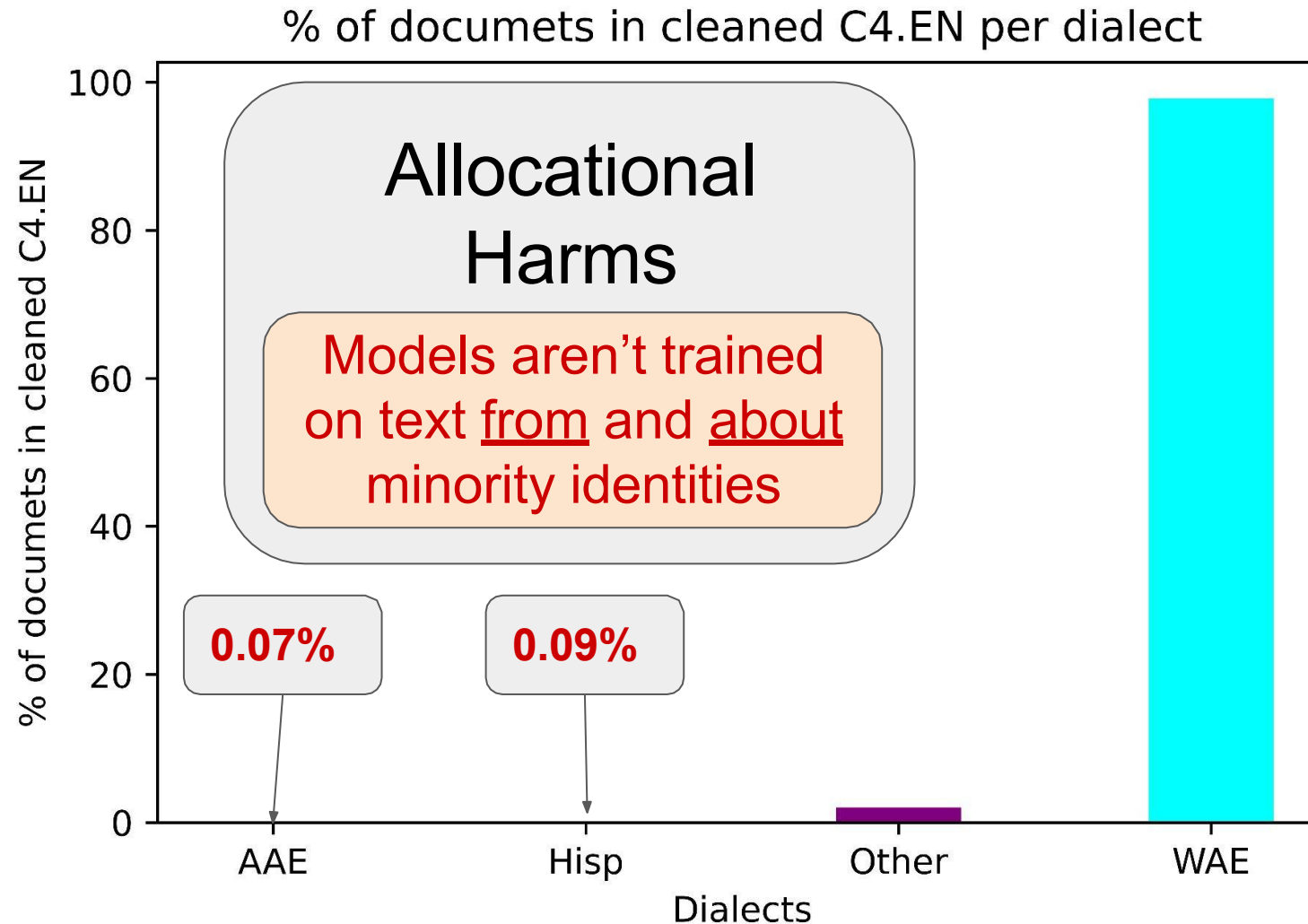
Content Removed from C4



Whose English is included?



Whose English is included?



Documenting Webtext Corpora: Recommendations

- ❑ Report metadata
 - Report domains the text is scraped from
 - Data collection process can lead to a different distribution of internet domains than one would expect
- ❑ Examine benchmark contamination
 - Support collecting data with the human-in-the-loop
 - To reduce contamination of future benchmarks
- ❑ Social biases and representational harms
 - Control the distributional biases
 - Select subdomains to use for training
 - Measurement of bias in each subdomain
- ❑ Excluded voices and identities
 - Avoid blacklist filtering when constructing datasets from web-crawled data
 - Some voices and identities might be excluded
 - Meaning of “bad” words heavily depends on the social context

Datasheets for datasets ([Gebru et al., 2018](#))

Datasheets for Datasets

TIMNIT GEBRU, Black in AI

JAMIE MORGENSTERN, University of Washington

BRIANA VECCHIONE, Cornell University

JENNIFER WORTMAN VAUGHAN, Microsoft Research

HANNA WALLACH, Microsoft Research

HAL DAUMÉ III, Microsoft Research; University of Maryland

KATE CRAWFORD, Microsoft Research

Does the dataset contain data that might be considered confidential (e.g., data that is protected by legal privilege or by doctor–patient confidentiality, data that includes the content of individuals’ non-public communications)? If so, please provide a description.

Unknown to the authors of the datasheet.

Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety? If so, please describe why.

Some movie reviews might contain moderately inappropriate or offensive language, but we do not expect this to be the norm.

Does the dataset identify any subpopulations (e.g., by age, gender)? If so, please describe how these subpopulations are identified and provide a description of their respective distributions within the dataset.

No.

Other Recommendations

Datasheets for datasets ([Gebru et al., 2018](#))

Data statements ([Bender & Friedman, 2018](#))

Data Statements for Natural Language Processing: Toward Mitigating System Bias and Enabling Better Science

Emily M. Bender
Department of Linguistics
University of Washington
ebender@uw.edu

Batya Friedman
The Information School
University of Washington
batya@uw.edu

C. SPEAKER DEMOGRAPHIC Sociolinguistics has found that variation (in pronunciation, prosody, word choice, and grammar) correlates with speaker demographic characteristics (Labov, 1966), as speakers use linguistic variation to construct and project identities (Eckert and Rickford, 2001). Transfer from native languages (L1) can affect the language produced by non-native (L2) speakers (Ellis, 1994, Ch. 8). A further important type of variation is disordered speech (e.g., dysarthria). Specifications include:

- Age
- Gender
- Race/ethnicity
- Native language
- Socioeconomic status
- Number of different speakers represented
- Presence of disordered speech

Other Recommendations

Datasheets for datasets ([Gebru et al., 2018](#))

Data statements ([Bender & Friedman, 2018](#))

Model cards ([Mitchell et al., 2018](#))

Dataset life cycle: <https://stanford-cs324.github.io/winter2022/lectures/data/>

Model Cards for Model Reporting

Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, Timnit Gebru
{mmitchellai,simonewu,andrewzaldivar,parkerbarnes,lucyvasserman,benhutch,espitzer,tgebru}@google.com
deborah.rajai@mail.utoronto.ca

- **Evaluation Data.** Details on the dataset(s) used for the quantitative analyses in the card.
 - Datasets
 - Motivation
 - Preprocessing
- **Training Data.** May not be possible to provide in practice. When possible, this section should mirror Evaluation Data. If such detail is not possible, minimal allowable information should be provided here, such as details of the distribution over various factors in the training datasets.
- **Quantitative Analyses**
 - Unitary results
 - Intersectional results
- **Ethical Considerations**
- **Caveats and Recommendations**

Limitations of the paper's analysis

Only examined some issues: location to report other issues

Tools used work disproportionately well for English:
generalization to other languages?

Limitations of the paper's analysis

Only examined some issues: location to report other issues

Tools used work disproportionately well for English:
generalization to other languages?

The Pile: An 800GB Dataset of Diverse Text for Language Modeling

Leo Gao

Stella Biderman

Sid Black

Laurence Golding

Travis Hoppe

Charles Foster

Jason Phang

Horace He

Anish Thite

Noa Nabeshima

Shawn Presser

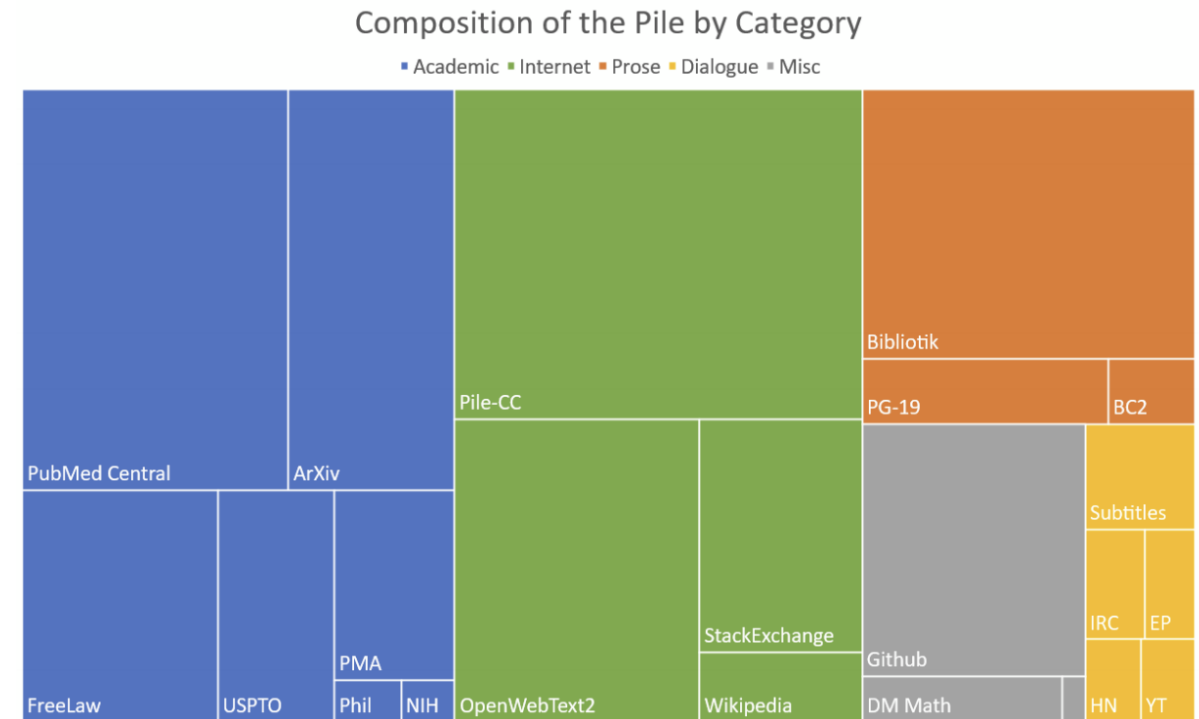
Connor Leahy

EleutherAI

December 2020

Motivation

- ❑ Dataset diversity leads to better downstream generalization (Rosset, 2019)
- ❑ LLMs are shown to acquire knowledge in new domain with relatively small training data
 - Large number of smaller high-quality datasets may improve cross domain knowledge + generalization



Deduplicating Training Data Makes Language Models Better

Katherine Lee^{*†}

Daphne Ippolito^{*†‡}

Andrew Nystrom[†]

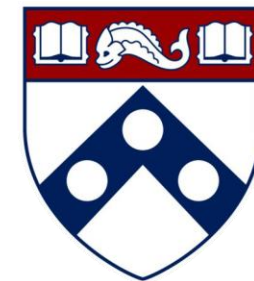
Chiyuan Zhang[†]

Douglas Eck[†]

Chris Callison-Burch[‡]

Nicholas Carlini[†]

Google Research



Penn
UNIVERSITY of PENNSYLVANIA

March 2022

Motivation

- ❑ Existing language modeling datasets contain many near-duplicate examples
- ❑ Long repetitive substrings (**%3 of C4 10M documents**)
- ❑ This encourages **memorization** and discourages **generalization**

Near duplicates

Example of near-duplicates in C4 dataset

Dataset	Example	Near-Duplicate Example
C4	Affordable and convenient holiday flights take off from your departure country, "Canada". From May 2019 to October 2019, Condor flights to your dream destination will be roughly 6 a week! Book your Halifax (YHZ) - Basel (BSL) flight now, and look forward to your "Switzerland" destination!	Affordable and convenient holiday flights take off from your departure country, "USA". From April 2019 to October 2019, Condor flights to your dream destination will be roughly 7 a week! Book your Maui Kahului (OGG) - Dubrovnik (DBV) flight now, and look forward to your "Croatia" destination!

Table is from [Lee et al, 2022](#)

Key Takeaways: Deduplicating paper

- ❑ Duplicates between the training and testing sets encourage the model to **memorize** the training data
- ❑ Deduplication does not harm, and sometimes improves model perplexity
- ❑ **Deduplication makes the training faster**

More detailed summary: <https://twitter.com/katherine1ee/status/1415496898241339400>

Key Takeaways

- ❑ A lot of data available on Web – Training on “all of it” not most efficient
- ❑ Filtering/curation needed, but results in biases Transparent
- ❑ Documentation needed:
 - **Data creators:** reflect on decisions, potential harms
 - **Data consumers:** know when dataset can/can't be used
- ❑ Curating non-web high quality datasets is promising (The Pile)
- ❑ When creating new dataset from the Web, remember that pretrained models may already have seen this data

CCNet: Extracting High Quality Monolingual Datasets from Web Crawl Data

**Guillaume Wenzek*, Marie-Anne Lachaux*, Alexis Conneau, Vishrav Chaudhary,
Francisco Guzmán, Armand Joulin, Edouard Grave**
Facebook AI

November 2019

mT5: A Massively Multilingual Pre-trained Text-to-Text Transformer

Linting Xue* **Noah Constant*** **Adam Roberts***
Mihir Kale **Rami Al-Rfou** **Aditya Siddhant** **Aditya Barua** **Colin Raffel**
Google Research

March 2021



Tokenization

Tokenization

- ❑ How do we represent an input text?
- ❑ So far in this class... we chop it up into words.
- ❑ ***Input text: students opened their books***
- ❑ ***Token IDs: 11 298 34 567***

This tokenization step requires an external *tokenizer* to detect word boundaries!

Word tokenization

- ❑ Not as simple as split on whitespace and punctuation...
 - *Mr. **O'Neill** thinks that the boys' stories about San Francisco **aren't** amusing.*
- ❑ Word tokenizers require lots of specialized rules about how to handle specific inputs
 - Check out **spaCy's tokenizers** (<https://spacy.io/>) and **Stanford Stanza's tokenizer** (<https://stanfordnlp.github.io/stanza/>).

Handling unknown words

- ❑ What happens when we encounter a word at test time that we've never seen in our training data?
 - With word level tokenization, we have no way of assigning an index to an unseen word!
 - This means we don't have a word embedding for that word and thus cannot process the input sequence
- ❑ **Solution:** Replace low-frequency words in training data with a special **<UNK>** token, use this token to handle unseen words at test time too
 - Why use **<UNK>** tokens during training?

Limitations of <UNK>

- We lose lots of information about texts with a lot of rare words / entities
 - The chapel is sometimes referred to as "Hen Gapel Lligwy" ("hen" being the Welsh word for "old" and "gapel" meaning "chapel").
 - The chapel is sometimes referred to as "Hen <UNK> <UNK>" ("hen" being the Welsh word for "old" and "<UNK>" meaning "chapel").
- Word-level tokenization treats different forms of the same word (e.g., “open”, “opened”, “opens”, “opening”, etc.) as separate types → separate embeddings for each.

This can be problematic especially when training over smaller datasets, why?

An alternative: character tokenization

- ❑ Small vocabulary, just the number of unique characters in the training data!
- ❑ However, you pay for this with longer input sequences. Why is this a problem for the models we've discussed?

Subword tokenization (2016)

- ❑ Developed for machine translation by Sennrich et al., ACL 2016
 - *“The main motivation behind this paper is that the translation of some words is transparent in that they are translatable by a competent translator even if they are novel to him or her, based on a translation of **known subword** units such as morphemes or phonemes.”*
- ❑ Later used in BERT, T5, RoBERTa, GPT, etc.
- ❑ Relies on a simple algorithm called byte pair encoding (BPE) (Gage, 1994)

Byte Pair Encoding (1)

- Form base vocabulary (all characters that occur in the training data)

word	frequency
hug	10
pug	5
pun	12
bun	4
hugs	5

- Base vocab: b, g, h, n, p, s, u

Example from https://huggingface.co/transformers/tokenizer_summary.html

Byte Pair Encoding (2)

- Now, count up the frequency of each character pair in the data, and choose the one that occurs most frequently

word	frequency
h+u+g	10
p+u+g	5
p+u+n	12
b+u+n	4
h+u+g+s	5

character pair	frequency
<i>ug</i>	20
<i>pu</i>	17
<i>un</i>	16
<i>hu</i>	15
<i>gs</i>	5

...

Example from https://huggingface.co/transformers/tokenizer_summary.html

Byte Pair Encoding (3)

- Now, choose the most common pair (ug) and then merge the characters together into one symbol. Add this new symbol to the vocabulary. Then, retokenize the data

word	frequency
<i>h+ug</i>	10
<i>p+ug</i>	5
<i>p+u+n</i>	12
<i>b+u+n</i>	4
<i>h+ug+s</i>	5

character pair	frequency
<i>un</i>	16
<i>h+ug</i>	15
<i>pu</i>	12
<i>p+ug</i>	5
<i>ug+s</i>	5

...

Example from https://huggingface.co/transformers/tokenizer_summary.html

Byte Pair Encoding (4)

- Keep repeating this process! This time we choose **un** to merge, next time we choose **h+ug**, etc.

word	frequency
<i>h+ug</i>	10
<i>p+ug</i>	5
<i>p+u+n</i>	12
<i>b+u+n</i>	4
<i>h+ug+s</i>	5

character pair	frequency
<i>un</i>	16
<i>h+ug</i>	15
<i>pu</i>	12
<i>p+ug</i>	5
<i>ug+s</i>	5

...

Example from https://huggingface.co/transformers/tokenizer_summary.html

Byte Pair Encoding (5)

- Eventually, after a *fixed number* of merge steps, we stop.

word	frequency
<i>hug</i>	10
<i>p+ug</i>	5
<i>p+un</i>	12
<i>b+un</i>	4
<i>hug + s</i>	5

new vocab: *b, g, h, n, p, s, u, ug, un, hug*

Byte Pair Encoding (6)

- ❑ To avoid <UNK>, all possible characters / symbols need to be included in the base vocab. This can be a lot if including all unicode characters (there are **~138K unicode symbols**)!
- ❑ GPT-2 uses bytes as the base vocabulary (size 256) and then applies BPE on top of this sequence (with some rules to prevent certain types of merges).
- ❑ Commonly have vocabulary sizes of 32K to 64K.

Other subword encoding schemes

❑ WordPiece (Schuster et al., ICASSP 2012)

- Merge by likelihood as measured by language model, not by frequency. Used in BERT.

❑ SentencePiece (Kudo et al., 2018)

- Can do subword tokenization without pretokenization (good for languages that don't always separate words w/ spaces), although pretokenization usually improves performance.
- Treats the input as a raw input stream.
- Includes the space (—) in the vocabulary.

အရှင်မင်းကြီး ရှေးဦးစွာ ကမ္ဘာအစ၌ဖြစ်သော မဟာသမတ မင်းကောင်း မင်းမြတ်တို့ ကျင့်တော်မူသော တရားကိုမြဲစွာ ကျင့်ဆောင်တည်တော်မူပါလော့။ အရှင်မင်းကြီး ခပ်သိမ်းသော ပြည်ထောင်မင်းတို့ကို အမျက်မလွန်စေခြင်း၊ ခပ်သိမ်းသော သူတို့ကို ကောင်းကြီးမြှောက်ခြင်း၊ ခပ်သိမ်းသောသူတို့အား ရင်တော်သားအမျှ ချစ်သနားခြင်း၊ ခပ်သိမ်းသောသူတို့ စီးပွားကိုစောင့်ခြင်း၊ ခပ်သိမ်းသောသူတို့အသက်ကို ကိုယ်တော်၏ အသက်ကဲ့သို့ ထင်မှတ်ခြင်းပြုပါလော့။ အသျှင်မင်းကြီး၊ လောဘ ဒေါသတို့၏ ပိတ်ပင်ခြင်းကို ကင်းစေ၍ အဝိဇ္ဇာမိုက်မဲစွာသော မောဟတရားကို လင်းတော် မူပါလော့။ အသျှင်မင်းကြီး ကျက်သရေရှိသော အမှုကိုသာပြုခြင်း၊ ကျက်သရေရှိသော စကားဆိုခြင်း၊ ကျက်သရေရှိသော လုံးလအကြံကိုသာ ကြံခြင်းတို့ကို ပြုတော်မူပါလော့။'

Burmese (Myanmar) language

Limitations of subwords

- Hard to apply to languages with agglutinative (e.g., زبان پیوندی یا چسبانشی) or non-concatenative (e.g., Arabic) morphology.

ک ت ب	k-t-b	“write” (root form)
کَتَبَ	kataba	“he wrote”
کَتَّبَ	kattaba	“he made (someone) write”
اِکْتَتَبَ	iktataba	“he signed up”

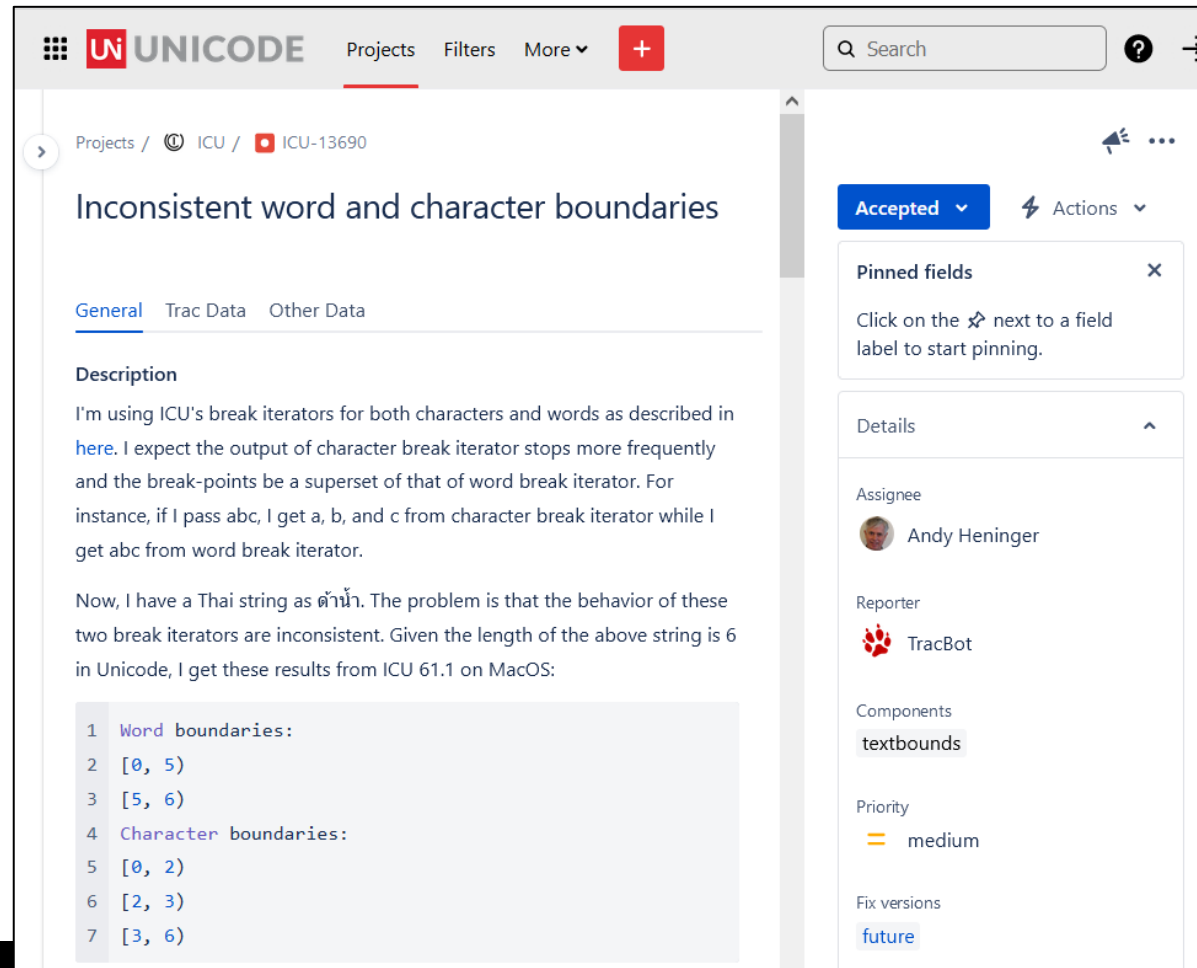
Table 1: Non-concatenative morphology in Arabic.⁴ The root contains only consonants; when conjugating, vowels, and sometimes consonants, are interleaved with the root. The root is not separable from its inflection via any contiguous split.

Limitations of subwords (cont'd)

- ❑ Pretokenization rules don't work on some languages (Thai and Chinese don't use spaces between words; Hawaiian uses punctuation as consonants.)

International Components for Unicode (ICU) is an open-source project of mature C/C++ and Java libraries for Unicode support, software internationalization, and software globalization.

Source: <https://unicode-org.atlassian.net/browse/ICU-13690>



The screenshot shows the Jira issue page for ICU-13690, titled "Inconsistent word and character boundaries". The issue is in the "Accepted" state. The description discusses the inconsistency between character and word break iterators in ICU, specifically mentioning a Thai string. A code block shows the results of the break iterators:

```
1 Word boundaries:
2 [0, 5)
3 [5, 6)
4 Character boundaries:
5 [0, 2)
6 [2, 3)
7 [3, 6)
```

The right sidebar shows the assignee as Andy Heninger, the reporter as TracBot, the component as textbounds, and the priority as medium. The fix version is set to future.

LLAMA Tokenizer (1)

LLAMA 1:

Tokenizer. We tokenize the data with the byte-pair encoding (BPE) algorithm (Sennrich et al., 2015), using the implementation from SentencePiece (Kudo and Richardson, 2018). Notably, we split all numbers into individual digits, and fallback to bytes to decompose unknown UTF-8 characters.

LLAMA 2:

Tokenizer. We use the same tokenizer as LLAMA 1; it employs a bytepair encoding (BPE) algorithm (Sennrich et al., 2016) using the implementation from SentencePiece (Kudo and Richardson, 2018). As with LLAMA 1, we split all numbers into individual digits and use bytes to decompose unknown UTF-8 characters. The total vocabulary size is 32k tokens.



LLAMA Tokenizer (2)



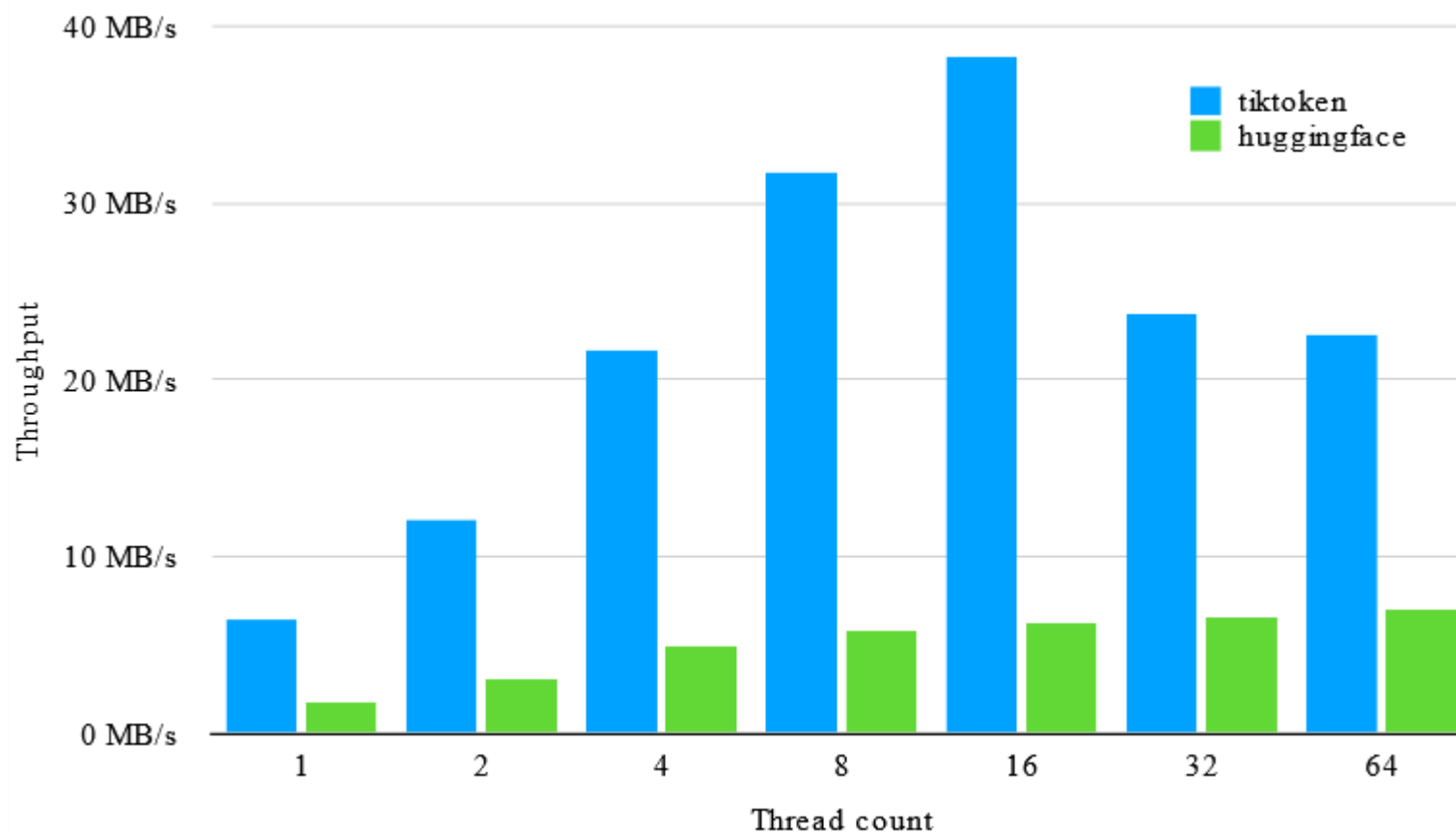
LLAMA 3:

We use a vocabulary with 128K tokens. Our token vocabulary combines 100K tokens from the tiktoken³ tokenizer with 28K additional tokens to better support non-English languages. Compared to the Llama 2 tokenizer, our new tokenizer improves compression rates on a sample of English data from 3.17 to 3.94 characters per token. This enables the model to “read” more text for the same amount of training compute. We also found that adding 28K tokens from select non-English languages improved both compression ratios and downstream performance, with no impact on English tokenization.



tiktoken

- ❑ A fast BPE tokeniser for use with OpenAI's models.
- ❑ Is 3-6x faster than a huggingface tokenizer.



Source: <https://github.com/openai/tiktoken>